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Deep learning-based time series forecasting for bearing remaining useful life: Recent advances, hybrid architectures, and targeted enhancements

Zhenyu Tang [a,b], Dalian Yang [a,b,*], Linrong Tan [c], Liying Zeng [c]

[a] School of Mechanical and Electrical Engineering, Hunan University of Science and Technology, Xiangtan, 411201, China
[b] Hunan Provincial Key Laboratory of Health Maintenance for Mechanical Equipment, Hunan University of Science and Technology, Xiangtan, 411201, China
[c] Xiangtan Industrial and Mining Electric Drive Vehicle Quality Inspection Center, Xiangtan, 411201, China

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ABSTRACT

This paper presents a systematic review of deep learning(DL)-based time series forecasting hybrid model for bearing RUL prediction by evaluating these studies from 2020 to 2025, addressing the gap in analyzing cross-model collaboration strategies. Firstly, the general workflow and modeling framework for remaining useful life (RUL) prediction were introduced. Secondly, a comparative analysis of time dependence capture mechanisms -including Recurrent Neural Networks(RNN) and their variants and Temporal Convolutional Network(TCN), and Transformer-was conducted. The findings reveal a shift from local sequential modeling to globally contextualized frameworks, enhanced by self-supervised learning and multimodal fusion. These developments underscore the importance of aligning model selection with specific task objectives and balancing multiple dimensions to achieve optimal performance in complex industrial scenarios. Thirdly, to address common challenges in RUL prediction, such as noisy data, sparse features, high model complexity, and limited interpretability, this paper explores performance optimization strategies from the perspectives of data quality, architectural design, and decision transparency. Finally, a comprehensive review was made of the current challenges: unclear applicability of attention, limited interpretability, deployment difficulties under resource constraints, difficulties in real-time prediction, and weak generalization in a data-scarce environment. To address these issues, future research should explore more trustworthy attention evaluation, interpretable architectures, lightweight design, continuous iteration for real-time prediction, and leveraging embedding other models to complement the lack of generalization. Overall, this work provides a structured perspective to support the development of the next generation of intelligent prediction systems.

RUL	Remaining Useful Life
DL	Deep Learning
RNN	Recurrent Neural Network
TCN	Temporal Convolutional Network
PHM	Prognostics and Health Management
EMD	Empirical Mode Decomposition
VMD	Variational Mode Decomposition
IMFs	Intrinsic Mode Functions
SVMD	Successive Variational Mode Decomposition
RMS	Root Mean Square
GAN	Generative Adversarial Networks
PCA	Principal Component Analysis
PCC	Pearson Correlation Coefficient
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
GRU	Gated Recurrent Unit

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MSE	Mean Squared Error
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
CRA	Cumulative Relative Accuracy
IFU	Information Fusion Unit
SA	Self-Attention
MHA	Multi-head attention
PAS	Partial Shrinkage Module
CPC	Contrastive Predictive Coding
PICP	Prediction Interval Coverage Probability
C-MAPSS	Commercial Modular Aero-Propulsion System Simulation
MMD	Maximum Mean Discrepancy
ES	Exponential Smoothing
WDRU	Weight-Decayed Recurrent Unit
GCN	Graph Convolutional Network

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* Corresponding author. School of Mechanical and Electrical Engineering, Hunan University of Science and Technology, Xiangtan, 411201, China.
E-mail address: ydlhnust@hnust.edu.cn (D. Yang).

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LNN	Liquid Neural Network
PDF	Probability Density Function
LLM	Large Language Model
FFVE	Factorized Feature Visualization and Embedding
VAE	Variational Autoencoder
PLCs	Programmable Logic Controllers
RCNN	Recurrent Convolutional Neural Network
PMCRCNN	Parallel Multichannel RCNN
TSA-LSTM	Tree Seed Algorithm Optimized LSTM
TDNN	Time Delay Neural Network
AWPME-GRU	Adaptive Wiener Process with Measurement Errors GRU
CMGRU	Consolidated Memory GRU
Attention-MGRU	Attention Mechanism GRU
TCNAAM	TCN with an Adaptive Activation Function and Attention Mechanism
TCN-RSA	TCN with Residual Self-attention Mechanism
TKEO-SSA-VMD-TCN	Teager-Kaiser energy operator and Sparrow Search Algorithm VMD and TCN
BiLSTM	Bidirectional LSTM
ASP-TCN	Adaptive Shrinkage Processing TCN
MP-TCN-Mixer	Multi-Scale Patch TCN-Mixer
SGGTCN	Separate Gating Mechanism and Global TCN
STFT	Short-time Fourier transform
TFT	Temporal Flow Transformer
MCFF	Melscale frequency cepstral coefficient
CDCT	Convolutional Dual-channel Transformer network
LECformer	Local Enhanced Channel Self-Attention based on Transformer
WTE-Trans	Weighted Time Embedding Transformer
DCTN-CBAM	Dual-channel Transformer network and Convolutional Block Attention Module
SAConvFormer	Spatial Attention-based Convolutional Transformer
ProgSViT	Prognostics Separable Vision Transformer
PMBCT	Probabilistic Multiscale Bayesian Convolutional Transformer
HA-ConvLSTM	Hybrid Attention-based ConvLSTM
MSDRSN	Multiscale Stacking Deep Residual Shrinkage network
CALAP	CNN-Attention-LSTM-Attention-Parallel
MS-DRNN	Multistream Deep RNN
TMLAM	Multi-channel model of transferable BiLSTM with an attention mechanism
DA-LSTM	Dual Attention LSTM
BLKAN	Bayesian large-kernel attention network
ACDBN	adaptive continuous deep belief networks
SHAP	SHapley Additive exPlanations

1. Introduction

In modern industrial systems, rotating machinery is extensively applied across various fields. Bearings serve as critical supporting components within rotating machinery, with their performance directly influencing equipment efficiency and operational safety (Peng et al., 2022). Many types of large scale equipment-such as generators, high-speed trains, and continuous production lines-are challenging to halt once operational, due to factors like high maintenance costs, stringent safety requirements, significant economic implications, and strong system interdependencies (Rezamand et al., 2020). During continuous equipment operation, failure to promptly monitor bearing health conditions may lead to severe accidents. For instance, in the offshore oil industry, up to 52 % of high-voltage induction motor failures have been attributed to bearing related issues (Silva and Cardoso, 2005). Similarly, bearing damages account for approximately 76 % of wind power gearbox failures (Evans, 2016). Additionally, statistical reports from Asea Brown Boveri Ltd., IEEE, and the Electric Power Research Institute indicate that bearing failures constitute a substantial proportion of the total damage in rotating machinery 51 %, 41 %, and 42 %, respectively (Chen et al., 2023). In light of these facts, research on predicting the RUL f bearings is particularly crucial. Accurate bearing RUL prediction not only reduces the frequency of unplanned downtime and associated maintenance costs but, more importantly, effectively prevents large-scale accidents (Nguyen et al., 2023; Xu and Saleh, 2021). Furthermore, accurate predictions provide a scientific decision-making basis for system design, production planning, and equipment maintenance, assisting enterprises in optimizing resource allocation and enhancing both operational efficiency and economic benefits (Kundu et al., 2021; Wu et al., 2024).

In recent years, RUL prediction technology has attracted widespread attention and achieved fruitful results in both academic research and practical industrial applications (Goswami and Nandan Rai, 2023). Currently, methods for bearing RUL prediction can be broadly categorized into physical model-based methods, statistical learning models, DL-based time series forecasting and its hybrid modeling methods.

The core principle of physical model-based RUL prediction methods is to establish a mathematical representation of bearing degradation mechanisms and failure modes, and then combine it with operational data collected in practice for in-depth analysis and calculation (Chen et al., 2023). A notable advantage of physical models lies in their strong interpretability, as they can explicitly reveal the underlying mechanisms of bearing degradation, making them essential tools for equipment design and optimization. When a physical model is constructed with high accuracy, its prediction results are typically more reliable. However, physical models heavily depend on prior knowledge, and constructing accurate physical models under unknown or complex operating conditions can be challenging, significantly limiting their applicability in practical engineering scenarios (Bienefeld et al., 2022).

To alleviate the human input in high-precision physical modeling, researchers have introduced statistical learning methods to achieve efficient modeling. Bearing RUL prediction methods based on statistical learning models excel at extracting complex features from large datasets, often uncovering patterns that are difficult for humans to detect. These methods perform effectively in scenarios with abundant data but unclear theoretical models, especially when dealing with unknown mechanisms or nonlinear systems. Common machine learning techniques for bearing RUL prediction include support vector machines (Shen and Yan, 2022), k-nearest neighbors (Chelmiah et al., 2022), and process regression (Cui et al., 2024a). However, statistical learning models approaches have limitations in handling high-dimensional or multi-source data, often necessitating complex feature extraction and dimensionality reduction procedures to achieve accurate predictions. This complicates their application in increasingly complex industrial scenarios (Y. Lu et al., 2024).Moreover, these methods heavily depend on the quantity and quality of the available data, ignoring the dynamic evolution inherent in time series data. Consequently, they may exhibit weak extrapolation capabilities, leading to potential overfitting or underfitting (Huang et al., 2024).

Bearing RUL prediction methods based on DL-based time series forecasting models have demonstrated outstanding high-dimensional and multi-source data processing capabilities, thanks to their advantages in modeling dynamic features in recent years. Typical examples include RNN, TCN, and Transformers (Lei et al., 2020). These models are particularly effective at capturing long range dependencies and dynamic evolution patterns in sequential data. They exhibit strong capabilities in analyzing trends, periodicities, seasonality, and other temporal characteristics. Furthermore, its effectively manage complex data originating from multiple variables and varying operational conditions. They are well-suited to the practical demands of increasingly complicated bearing operational scenarios and the growing diversity in data acquisition types, consistently demonstrating strong performance in dynamically evolving prediction tasks (Li et al., 2023, 2024). However, the parameters of DL-based time series forecasting models generally lack clear physical interpretations, making it difficult to provide credible explanations for prediction outcomes (Mian et al., 2024).

To overcome the aforementioned limitations associated with physical models, statistical learning models, and standard DL-based time series forecasting models, researchers have proposed hybrid modeling strategies based on time series approaches. The core concept involves integrating two or more network architectures, leveraging their complementary advantages to construct a novel RUL prediction framework

that simultaneously achieves high prediction performance and strong interpretability (Kim et al., 2024). For instance, combining TCN with attention mechanism(Wang et al., 2023), Traditional physical degradation models are combined with DL-based time series forecasting models (Zhu et al., 2024). Such hybrid modeling strategies based on time series have gradually emerged as a research hotspot within the bearing RUL prediction field, leading to numerous innovative studies and constructive outcomes in recent years. Although DL-based time series forecasting models and their hybrid models have significantly promoted the development of bearing RUL prediction, there are still three major gaps in the field.

(1) There is a lack of systematic analysis of the performance differences of different DL-based time series forecasting models in RUL prediction.
(2) The mechanism of module synergy and advantage complementarity in hybrid model has not been thoroughly explained.
(3) There is a lack of systematic reviews to summarize the challenges in this field and point out the key research directions in the future.

The remainder of this paper is structured as follows: Section 2 elaborates on the methodology of this review. Section 3 introduces the general procedures for RUL prediction. Section 4 focuses on DL-based time series forecasting models, systematically analyzing their theoretical foundations, main characteristics, suitable application scenarios, and practical performances in RUL prediction tasks. Section 5 provides an in-depth exploration of the advantages and applicability of hybrid modeling strategies, comparing their predictive performance and their ability to address specific problems, while summarizing solutions for various critical issues. Section 6 identifies major challenges and future development trends based on practical industrial needs, offering insights and guidance for future research. Finally, Section 7 presents the conclusion of the paper.

2. Review method

To systematically and objectively review the research progress of DL-based time series forecasting hybrid models in the field of bearing RUL prediction, this study conducted a literature search using the Web of Science database. The search focused on relevant literature published between January 2020 and July 2025 that applied time series models for bearing RUL prediction. The search keywords were set as "bearing AND remaining useful life AND (time series OR time series analysis OR temporal modeling OR sequence modeling)". A total of 252 entries were retrieved (https://webofscience.clarivate.cn/wos/woscc/summary/e9b03924-cecf-47f4-91ba-4c4eeba466b5-0151e07266/relevance/1). Subsequently, the retrieved literature was screened and reviewed to exclude irrelevant articles, resulting in 241 relevant articles. Among these, 140 articles focused on hybrid DL-based time series forecasting models, and only one was a review article (Liu et al., 2025).The annual distribution and yearly citation counts of the retrieved literature are shown in Fig. 1.

From Fig. 1, it can be observed that between 2020 and 2024, the number of publications related to bearing RUL prediction based on time series models showed a steady upward trend, reaching its peak in 2024 with a total of 77 publications. As of July 2025, 23 relevant papers have already been published, accumulating 838 citations, reflecting growing attention toward bearing RUL prediction research using time series modeling methods. Moreover, research on hybrid time series forecasting models has increased notably in recent years, particularly showing significant growth in 2024 and 2025, indicating that hybrid modeling approaches have become a current research hotspot. Additionally, to highlight the contributions of this study, we compared its key contributions with previously published review articles, as shown in Table 1.

Based on Table 1, the literature (Liu et al., 2025) primarily emphasizes the significance of improving the predictive performance of deep learning models through multi-source data fusion, providing an overall description within the framework of PHM for bearing health monitoring. However, the discussion regarding model optimization strategies and hybrid models in that study is relatively limited, and it lacks a systematic and comprehensive summary and analysis of the latest advancements and cutting-edge applications of hybrid DL-based time series forecasting models in bearing RUL prediction.

Given this gap, the present paper aims to systematically review the research status and development trends of hybrid DL-based time series forecasting models for bearing RUL prediction tasks. It focuses on analyzing their applicability and limitations in practical industrial scenarios, summarizing feasible solutions proposed in existing studies to address core issues in RUL prediction. Furthermore, through systematic literature retrieval and comparative analysis, this study identifies key factors influencing model performance and explores potential optimization strategies aligned with industrial application requirements. The main contributions of this paper are as follows.

(1) Comprehensively analyzed the unique advantages of various DL-based time series forecasting models in RUL prediction tasks, emphasizing their differentiated performances in handling temporal dependencies, feature extraction capabilities, and

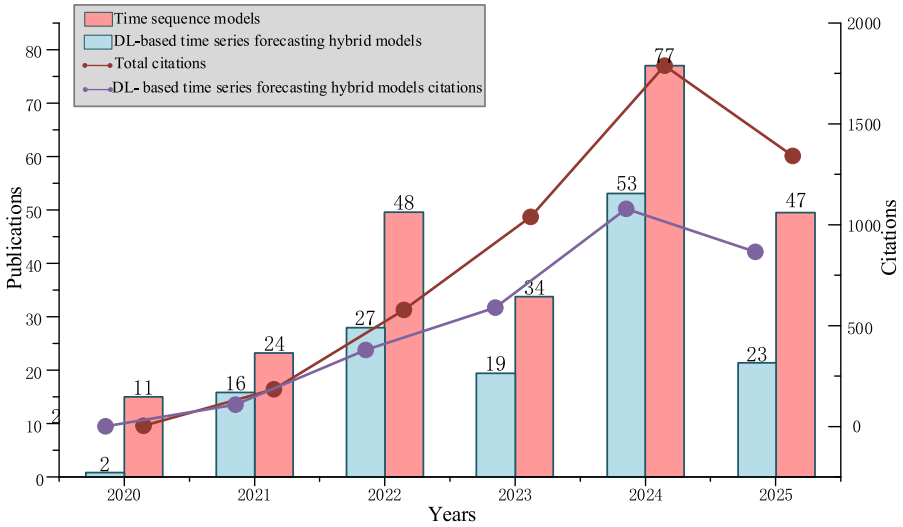


Fig. 1. Number of publications and citations in Web of Science between January 2020 and July 2025.

Table 1
Comparison with existing reviews.

Comparing dimensions	Liu et al. (Liu et al., 2025)	This paper	Tackling overlooked aspects
Research Objectives	A macro overview of their sequence modeling mechanism and long distance dependence ability is given.	A comparative framework of RNN, TCN and Transformer is constructed to comb the evolution of their long dependence modeling mechanism.	At the mechanism level, the modeling differences and advantages of three kinds of deep temporal networks in RUL tasks are compared.
Classification system	The task flow classification from anomaly detection to HI construction to degradation modeling to RUL prediction is introduced	Taking the model architecture and coordination mode as the main line, the serial-parallel connection and attention coupling of the hybrid model are expounded	Supplement the systematic classification of the model architecture hierarchy to facilitate engineering applications and secondary development
Hybrid model coordination mechanism	The idea of hybrid modeling is mentioned, but the concrete realization is not given	demonstrate the advantages of hybrid models under challenges such as noise, long-range dependence, and interpretability	This paper illustrates that the closed loop from concept to operational paradigm fills the gap where the literature stays at the conceptual level
Comparison of methods	The quantitative differences of RUL prediction accuracy, inference efficiency and interpretability among different temporal networks were not compared	Experimental cases and quantitative indicators are provided to systematically reveal the advantages and disadvantages of each model	Comparing the literature method gap, from fuzzy description to quantitative contrast

computational efficiency. Additionally, this study assessed the applicability of these DL-based time series forecasting models in industrial scenarios, providing theoretical references and practical guidelines for researchers in this field.

(2) Conducted an in-depth investigation from the perspectives of complementary network structures, module collaboration mechanisms, and specific problem-solving capabilities, thoroughly analyzing the models' contributions to optimizing bearing RUL predictions and their practical application value. Moreover, this review systematically summarized how researchers leveraged the complementary strengths of different network models, significantly improving prediction accuracy and robustness by enhancing feature extraction methods and optimizing degradation trend modeling.

(3) Outlined the current challenges faced by hybrid DL-based time series forecasting models in RUL prediction and provided insights into potential future research directions.

3. Bearing RUL prediction process based on DL-based time series forecasting model

Under the rapid development of modern industry, bearings play a crucial role in ensuring reliable operation of machinery and equipment (Wang et al., 2020b). Bearings not only suppress relative motion but also effectively reduce friction between moving components. However, during their continuous service, bearings inevitably experience various types of faults, potentially causing equipment failures, economic losses, or even catastrophic accidents (Behera and Misra, 2023). Hence, it becomes particularly critical to predict future degradation trends and estimate the RUL of bearings (Wang et al., 2024). RUL prediction is typically formulated as a regression task, modeling the degradation process of equipment to estimate the remaining time from the current state to failure, thereby providing support for the health management of industrial equipment. The existing RUL prediction methodologies generally consist of five key steps: data acquisition, data processing, feature extraction, model training, model evaluation, and prediction (Gong et al., 2022; Hachem et al., 2024; Wang et al., 2023), as illustrated in Fig. 2. This section outlines the general framework and critical steps involved in RUL prediction.

3.1. Data acquisition

Data acquisition forms the foundation for RUL prediction and involves collecting raw signals that reflect the operational status of bearings. Common data sources include vibration signals, acoustic signals, temperature measurements, rotational speed, and multi-sensor fusion signals among others. Specifically, vibration acceleration signals from rolling bearings contain rich and useful information, and desired vibration signals can be obtained from the target bearings via sensor devices (Pang et al., 2023). The quality and resolution of the collected data are critical to the success of subsequent analytical research.

In current research on rolling bearing RUL prediction, researchers commonly utilize publicly available bearing degradation datasets from various experimental platforms as input data to validate the effectiveness and robustness of proposed prediction methods. Among the most frequently used datasets are the XJTU-SY Bearing Dataset and the IEEE PHM 2012 Challenge Dataset. The XJTU-SY Bearing Dataset is specifically designed for research on bearing degradation prediction, providing complete lifetime operational data of bearings under multiple operating conditions. It effectively supports the construction and validation of RUL prediction models for bearings under different degradation modes. The dataset was collected using a rolling bearing test rig, as illustrated in Fig. 3(a) (Wang et al., 2020a). The experimental platform consists of an AC motor, a motor speed controller, a support shaft, two supporting bearings, and a hydraulic loading system (Wang et al., 2019). This test rig can perform accelerated degradation tests under various conditions, providing vibration signal data from 15 LDK UER204 ball bearings throughout their lifecycle, under three different conditions (Wang et al., 2021): 2100 r/min with 12 kN load, 2250 r/min with 11 kN load, and 2400 r/min with 10 kN load. Data was collected every minute at a sampling frequency of 25.6 kHz, with each frame lasting 1.28 s, yielding 32,768 data points. Examples of vibration signals from this dataset are shown in Fig. 4(a) (Gao et al., 2024).The IEEE PHM 2012 Challenge Dataset was obtained from the PRONOSTIA platform and contains accelerated degradation signals from rolling bearings under different operational conditions (Nectoux et al., 2012).The PRONOSTIA platform, shown in Fig. 3(b), is an experimental facility specifically designed for research on bearing performance degradation and RUL prediction (Li et al., 2024). It provides realistic experimental data representing degradation patterns of ball bearings throughout their entire lifespan. The data was collected every 10 s, with each frame lasting 0.1 s, sampled at 25.6 kHz, resulting in 2560 data points per frame, as illustrated in Fig. 4(b). To evaluate the generalization performance of prediction models, researchers have also adopted datasets collected from other experimental platforms, such as the C-MAPSS dataset provided by NASA. This dataset simulates turbofan engine degradation through a commercial modular aerospace propulsion system (Saxena et al., n.d.). It comprises four subsets representing different failure modes and operating conditions (Xu et al., 2022). Furthermore, to validate model performance in scenarios with limited or incomplete data, several researchers have utilized publicly available bearing fault datasets such as the IMS Bearing Dataset (Von Hahn and Mechefske, 2022) and the

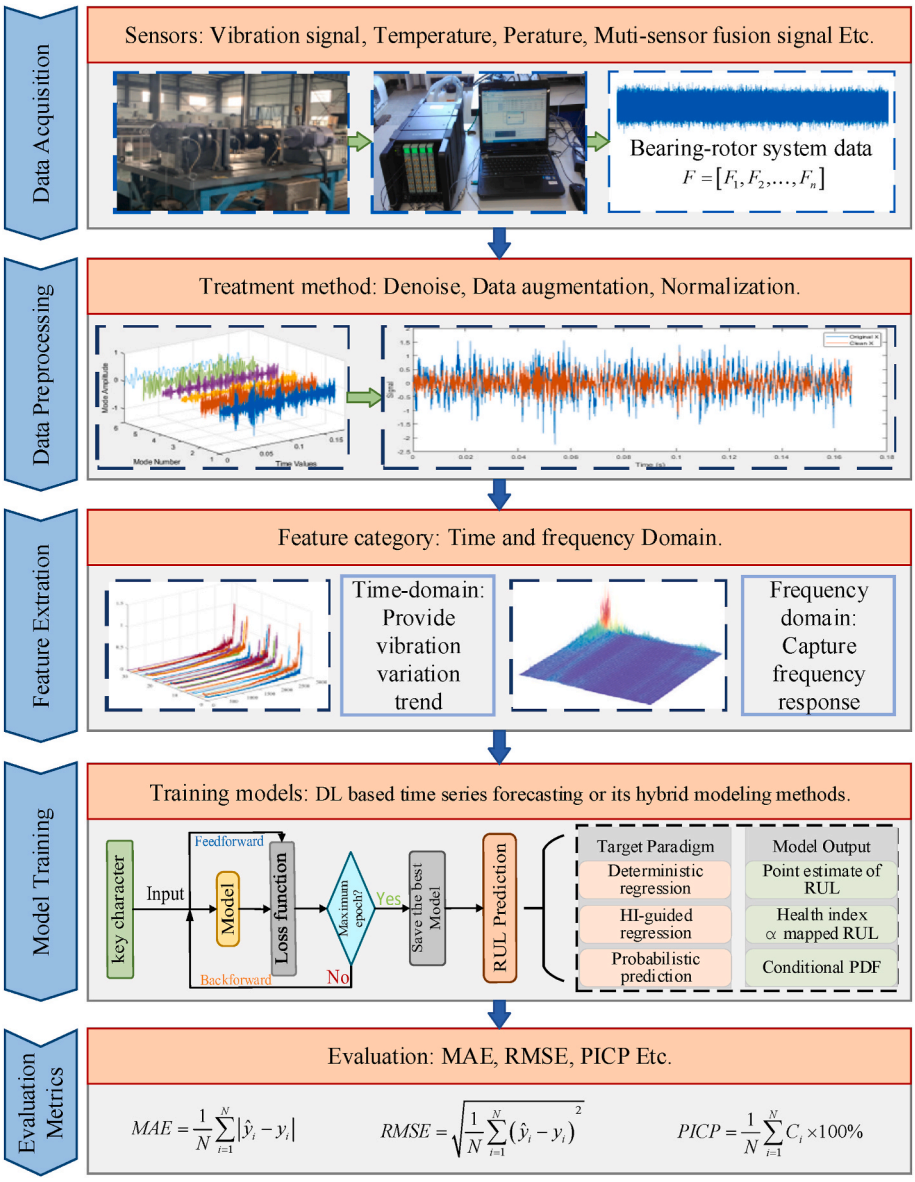


Fig. 2. General process of RUL prediction.

CWRU Bearing Dataset (Cheng et al., 2021).

In addition, some researchers use multi-sensor fusion data to predict the remaining service life of bearings. These studies do not rely on a single acceleration signal but rather on synchronous heterogeneous signals. In the field of wind turbines, CVC-Net integrates ultrasonic and vibration flow, reducing the misclassification of early bearing faults by one third (Li et al., 2024). A decision-level multimodal scheme that aggregates the singular spectral features of three sensors further enhances the prediction robustness under high-noise conditions. Recently, a dual-attention multi-scale fusion network demonstrated that by jointly learning the correlation between the spectra and time of four synchronous channels, the reliable prediction range can be extended by up to 20 %(Yang et al., 2025). These advancements confirm that sensor-level complementarity is becoming a key asset for RUL modeling and illustrate a clear trend towards publicly sharing multimodal datasets to facilitate reproducible research and benchmark future algorithms.

3.2. Data preprocessing

The theoretical accuracy of any bearing-RUL model is bounded by the quality of its input signals (Yao et al., 2024d). In practice, sensor limitations and complex operating conditions introduce record-length inconsistencies, parameter drift, noise and missing segments (Xie et al., 2024). Consequently, systematic data preprocessing-including denoising, data augmentation, and normalization-is indispensable before modelling can begin (She et al., 2024).

Denoising methods include Transform-based Methods (Huang et al., 2024), Decomposition-based Methods(Liu et al., 2022), Adaptive Filtering Methods(Szabo et al., 2024), Data-driven or Learning-based Methods(Lu et al., 2024), Statistical and Empirical Methods(Yu et al., 2025).

Data augmentation techniques include oversampling(Bao et al., 2020; Dai et al., 2023), downsampling, sample repair(Liu et al., 2022; Yang et al., 2024a), GANs, VAEs, hybrid generative models(Shi et al., 2023; Verstraete et al., 2020; Wei et al., 2025).

Normalization Approaches include min–max normalization, z-score normalization(Li et al., 2024; Maurya and Verma, 2023).

In summary, high-quality preprocessing is a prerequisite for reliable RUL modeling, especially under noisy and incomplete conditions. Denoising enhances signal clarity, augmentation expands training

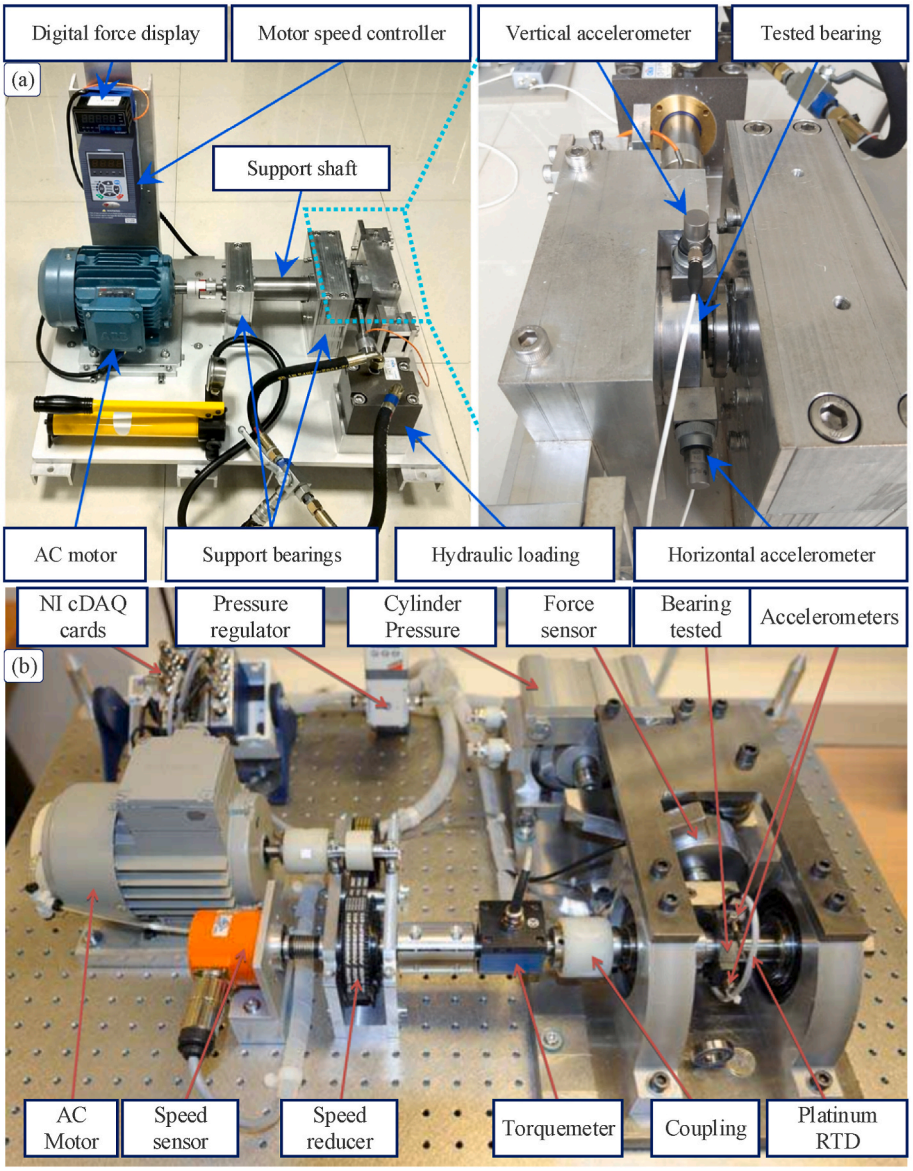


Fig. 3. Testbed of rolling element bearings. (a) XJTU-SY(B. Wang et al., 2020a); (b)PHM2012(Nectoux et al., 2012).

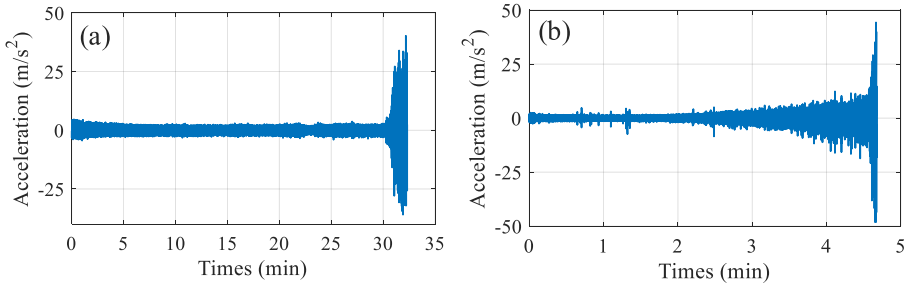


Fig. 4. Examples of visualization of vibration signals. (a) XJTU-XY; (b)PHM 2012.

diversity, and normalization standardizes input features. The synergistic application of these methods forms a robust foundation for data-driven prognostic modeling.

3.3. Feature extraction and model training

During the feature extraction phase, degradation features of bearings are obtained either through signal analysis methods or automatically via deep learning techniques (Yao et al., 2024c). Traditional signal analysis approaches-such as time-domain, frequency-domain, and time-frequency

domain methods-are widely used in the field of RUL prediction, as they can extract both low-level features and intuitive, concise representations (Liang et al., 2023). Among features derived from the time, frequency, and time-frequency domains, those representing amplitude characteristics include RMS, frequency band energy, peak value, dominant frequency, and kurtosis factor (Xu et al., 2021). Features that describe signal trends and variations include mean value, spectral entropy, skewness, kurtosis, and wavelet coefficients. Local features of signals are often captured using skewness, kurtosis, and wavelet coefficients(Chen et al., 2020). Indicators that reflect the energy characteristics of vibration signals mainly include RMS (Hou et al., 2024), wavelet energy spectrum, and short-time Fourier transform energy spectrum (Wang et al., 2016).

Although the features extracted from vibration signals can effectively reflect the intrinsic information of the bearing degradation process, not all features exhibit the same level of sensitivity to different fault types. Therefore, in rolling bearing RUL prediction tasks, it is essential to perform targeted selection and transformation of the extracted features. Dimensionality reduction techniques are often employed to extract key features that are both sensitive and highly discriminative under various operating conditions. This helps avoid the inclusion of redundant high-dimensional noise, thereby reducing the computational complexity of the model, improving prediction accuracy, and enhancing the interpretability of the results (Cao et al., 2024; Ma et al., 2024). In addition to dimensionality reduction for handling excessive feature quantities, commonly used feature selection methods include PCA (Jiang et al., 2020), PCC (Mao et al., 2020), and L1 regularization (Ma et al., 2024). To improve training efficiency, redundant features should be eliminated, and the most representative features or feature combinations that effectively capture the bearing degradation process should be selected.

However, although effective feature selection and dimensionality reduction methods can eliminate redundant features and retain key features that are highly discriminative and sensitive to degradation states-thus enhancing model prediction performance (Miao and Yu, 2021), -manually designed features often require task-specific network architectures and substantial prior knowledge. Moreover, such features tend to be less responsive to the degradation process and may fail to provide sufficient useful information (Wang et al., 2019) With the rise of deep learning, an increasing number of researchers have adopted automated feature extraction methods. These methods enable networks to automatically learn meaningful features directly from raw data through training (Zhang et al., 2020).An et al. (2020) proposed a hybrid approach combining CNN and LSTM networks, where CNN are used for local feature extraction and dimensionality reduction, and an LSTM layer is added to the front of the CNN to capture more abstract representations and encode temporal information. Researchers have also found that combining signal analysis techniques with DL-based time series forecasting models can yield more effective feature extraction. Signal analysis methods help extract low-level, intuitive features, while DL-based time series forecasting models are better suited for capturing deeper, more abstract patterns. The fusion of features across different levels compensates for the limitations of using single-layer features. For instance, Wang et al. (2022) extracted features using time-domain, frequency-domain, and time-frequency domain methods, while simultaneously employing bidirectional LSTM to capture deep-level features. By leveraging the complementarity between low-dimensional and high-dimensional features, they significantly improved RUL prediction performance. Therefore, feature extraction strategies should be selected comprehensively and appropriately, considering the specific requirements of the task, the input needs of the model, and the characteristics of the available data.

In addition, During the Model-Training stage illustrated in Fig. 2, raw time-frequency features and operating-condition signals are first passed through a key-character filter that keeps only those channels most sensitive to bearing degradation (Liu et al., 2023). The refined input then flows through a candidate deep network whose output format is selected according to the overall prediction paradigm: it can deliver a single RUL point estimate, a latent health-index that is later converted to RUL, or an entire probability distribution that captures predictive uncertainty. The network parameters are updated by back-propagating an appropriate loss-mean error metrics for point estimates, a hybrid loss for health-index mapping, or a likelihood-based objective for probabilistic prediction (Tao Zuo et al., 2023). If an early-stopping threshold or the predefined epoch limit is reached, the best checkpoint is saved for deployment.

3.4. Model evaluation and prediction

The primary objective of constructing rolling bearing RUL prediction models is to minimize human intervention, enabling users to effectively apply these models with minimal reliance on expert prior knowledge. This not only enhances the generalizability and applicability of the models but also opens up greater possibilities for innovation and advancement in this research field (Yang et al., 2024). Against this backdrop, it becomes particularly important to objectively and effectively evaluate the performance and reliability of prediction models. Typically, model evaluation involves a comprehensive analysis across several dimensions, including predictive performance, generalization capability, and robustness (Wang et al., 2022). Since RUL prediction is essentially a regression task, commonly used performance metrics include MSE (Guo et al., 2024c), MAE (Shao et al., 2024), Coefficient of Determination (Wei et al., 2023), and RMSE (Jiang et al., 2020). Among them, MSE and RMSE measure the magnitude of the deviation between predicted values and ground truth, MAE reflects the average absolute difference between predictions and actual values, while R^2 indicates the extent to which the model explains the variance in the observed data (Kim et al., 2024a; Yu and Zhang, 2023). Furthermore, the generalization ability of a model refers to its performance on unseen data not involved in training, and is typically validated through techniques such as cross-validation (Chen et al., 2024; Xiang et al., 2024). Robustness, on the other hand, denotes the model's ability to maintain stable predictive performance in the presence of noise, missing values, or other data quality issues (Liao et al., 2024). Therefore, the evaluation of bearing RUL prediction models should not be limited to a single metric. A comprehensive assessment must consider predictive accuracy, generalization capability, robustness, and the computational cost of training and inference. In practical industrial applications, it is especially important to strike a balance between model performance and resource consumption to ensure efficient deployment and real-world applicability.

To substantially enhance the generalization ability and robustness of models, recent studies typically focus on the following three complementary directions. The first is the enhancement at the data level, such as synthetic fault injection based on GAN or VAE (Wang et al., 2024), and cross-machine transfer learning through adversarial domain adaptation to align simulations and real signals(Du et al., 2024; Shang et al., 2024); The second is regularization at the model level, introducing residual blocks guided by physical mechanisms(Yan et al., 2023; S. Yang et al., 2024b), the spectrum-time domain attention mask, and the multi-task head that predicts the health index and RUL simultaneously, to inject stronger structural priors into the network and suppress overfitting (Wang et al., 2024). The third is the defense measures oriented towards robustness. During the training process, noise perturbations or channel masks are added. Bayesian or Evidential layers are used to quantify the uncertainty of prediction, and lightweight adaptation is carried out in the inference stage to compensate for sensor drift and missing data, without the need to completely retrain the model(Xuan et al., 2023).

In summary, genuine progress on bearing-RUL prediction arises when data augmentation, physics-aware architecture design, disciplined optimization, and uncertainty quantification are orchestrated as a unified pipeline; treating any one facet in isolation yields only incremental gains, whereas their systematic integration transforms accuracy into

reliability under real-world perturbations.

4. DL-based time series forecasting models for RUL prediction: overview

4.1. RNNs and its variants model

4.1.1. RNNs

Traditional machine learning methods rely heavily on manually designed features and threshold settings, making them poorly suited to adapt to distributional shifts in data under complex operating conditions (Han et al., 2021). When dealing with sequential or time series data, conventional feedforward neural networks are inadequate for learning temporal dependencies or making sequence-based predictions (Zhang et al., 2022). These networks are designed for independently distributed data points and operate with a single input and output, limiting their effectiveness in time-dependent scenarios. To address this limitation, researchers proposed the RNN model(Rumelhart et al., 1986; Zaremba et al., 2015). RNNs have demonstrated significant advantages in the field of bearing RUL prediction due to their ability to handle temporal sequences and model dynamic dependencies inherent in degradation processes (Mienye et al., 2024). The model computes task-specific loss and uses Back Propagation Through Time to determine gradients. By calculating the error from the input to the output layer, the model updates its parameters accordingly. During parameter sharing across time steps, BPTT aggregates the error across all time steps, enabling RNNs to effectively capture sequential patterns. The architecture of a typical RNN is illustrated in Fig. 5 (Guo et al., 2017; Shang et al., 2022).

RNNs are highly flexible, with the ability to adapt the lengths of input and output sequences according to the specific requirements of the prediction task (Rumelhart et al., 1986). During operation, an RNN processes sequential data starting from time step $t = 1$, generating a series of hidden states $\{h_1, h_2, \ldots\}$and corresponding outputs $\{y_1, y_2, \ldots\}$ that capture the temporal dependencies within the sequence. as shown in Equations (1) and (2).

$$h_t = f(Wh_{t-1} + Ux_t + b_h) \tag{1}$$

$$y_t = Vh_t + b_y \tag{2}$$

Here, W denotes the recurrent weight matrix of the hidden layer, U represents the input-to-hidden weight matrix, b_h is the bias term of the hidden layer. The activation function $f(\cdot)$ is typically a nonlinear function such as tanh or ReLU.

Although RNNs demonstrate strong capabilities in processing time series data, they also suffer from several inherent limitations. When dealing with long sequences, RNNs are prone to issues during backpropagation - specifically, the repeated multiplication of weight matrices can cause gradients to exponentially vanish or explode, leading to gradient vanishing or gradient explosion problems. In addition, RNNs process sequential data one time step at a time, resulting in memory consumption that increases proportionally with sequence length. This imposes a high demand on computational resources and leads to slower training speeds (Li et al., 2020). Even though RNNs are theoretically capable of learning from long sequences, in practice, the hidden states tend to gradually forget earlier information over time, making it difficult to maintain sensitivity to long-range dependencies-a major challenge for this network architecture (Lei et al., 2008). To address these issues, researchers have proposed several improvements. One effective approach is the integration of attention mechanisms with RNNs. Specifically, causal attention mechanisms have been designed to accommodate the non-stationary nature of industrial signals by adaptively focusing on critical time steps, significantly enhancing the model's ability to capture long-term dependencies. For example, Chen et al. (2020) utilized an encoder–decoder framework combined with an attention mechanism to dynamically allocate weights to key stages in the degradation process. Zhang et al. (2022) further improved long-sequence modeling performance by incorporating both channel and spatial attention modules into the RNN framework. To mitigate the gradient vanishing and explosion problems, researchers have also modified the conventional RNN structure by introducing gated mechanisms, such as LSTM and GRU, which have proven effective in stabilizing gradient flow during training.

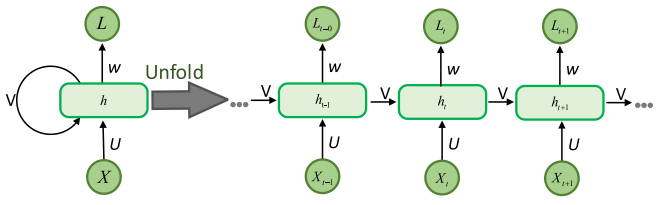


Fig. 5. Traditional RNN model.

4.1.2. LSTM

To overcome the limitations of RNNs in modeling long sequences-such as inadequate long-term dependency capture and the risks of gradient explosion or vanishing-researchers introduced a gating mechanism into the RNN architecture, resulting in the development of the LSTM (Hochreiter and Schmidhuber, 1997). By incorporating three types of gates-namely, the forget gate, input gate, and output gate-LSTM effectively improves the ability to retain and utilize historical information, significantly enhancing its capacity to model long-term dependencies. The architecture of LSTM is illustrated in Fig. 6. Owing to its gated structure, LSTM has demonstrated outstanding performance in time series modeling tasks and has been widely adopted in bearing RUL prediction, becoming one of the key technologies in the field of PHM (Huang et al., 2007; Sarlin, 2013).

LSTM preserves critical degradation information over long periods through its memory cell, enabling the model to learn and establish long-range dependencies from the initial stages of degradation to final failure (Ma and Mao, 2021). It not only matches, but often surpasses the capabilities of traditional RNNs, while effectively mitigating issues such as gradient vanishing and explosion (Hu et al., 2022).The mathematical formulation of the LSTM is expressed as follows:

$$f_t^l = \sigma\left(W_f^l h_{t-1}^l + U_f^l x_t^l + b_f^l\right) \tag{3}$$

$$i_t^l = \sigma\left(W_i^l h_{t-1}^l + U_i^l x_t^l + b_i^l\right) \tag{4}$$

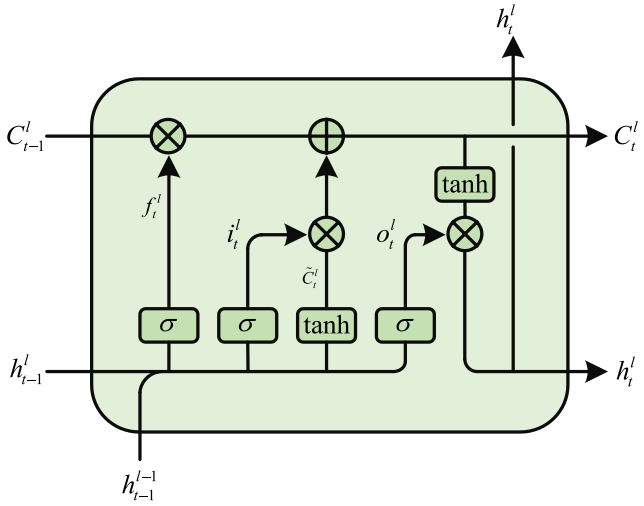


Fig. 6. LSTM framework.

$$\tilde{c}_t^l = \tanh\left(W_i^l h_{t-1}^l + U_i^l x_t^l + b_i^l\right) \tag{5}$$

$$c_t^l = f_t^l \odot c_{t-1}^l + i_t^l \odot \tilde{c}_t^l \tag{6}$$

$$o_t^l = \sigma\left(W_o^l h_{t-1}^l + U_o^l x_t^l + b_o^l\right) \tag{7}$$

$$h_t^l = o_t^l \odot \tanh\left(c_t^l\right) \tag{8}$$

Here, f_t^l is the forget gate, which determines how much of the previous cell state should be retained; i_t^l is the input gate, $\tilde{c}_t^l$ is the candidate cell state-together, they control the incorporation of new information at the current time step. c_t^l represents the updated cell state. The output gate o_t^l determines how much of the cell state information is passed to the next hidden state. h_t^l denotes the updated hidden state, $\sigma(\cdot)$ represents the sigmoid activation function, tanh $(\cdot)$ denotes the hyperbolic tangent function and $\odot$ indicates the element-wise product.

However, traditional LSTM models still face limitations when applied to scenarios involving multi-sensor data fusion, cross-domain generalization, and small-sample learning. To address these challenges, Li et al. (2022) proposed a two-dimensional LSTM, which integrates horizontal vibration signals and numerical vibration signals collected from vibration sensors. This approach enables the extraction of spatiotemporal degradation features while performing data augmentation, allowing the model to learn consistent degradation patterns from multiple signal sources. Cai et al. (2024) employed a bidirectional LSTM combined with a degradation detection strategy, achieving a 40 % reduction in prediction error compared to traditional LSTM models.

4.1.3. GRU

Although LSTM-based recurrent neural networks demonstrate strong performance in capturing long-term dependencies within time series data, their complex gating mechanisms significantly increase the number of model parameters and computational overhead. This becomes particularly problematic when handling long sequences, often leading to reduced training efficiency. To address this limitation, researchers proposed the GRU, which achieves more efficient training in many practical applications. As a variant of the RNN, GRU simplifies the gating structure of LSTM by merging the forget gate and input gate into a single update gate, resulting in only two gates: the update gate and the reset gate, as illustrated in Fig. 7. This simplification reduces model complexity while maintaining strong temporal modeling capabilities (Wang et al., 2022).

Compared to the traditional LSTM, the GRU architecture is more compact. Its core components are the update gate and reset gate, and it maintains a single hidden state vector. The mathematical formulation of the GRU is expressed as follows:

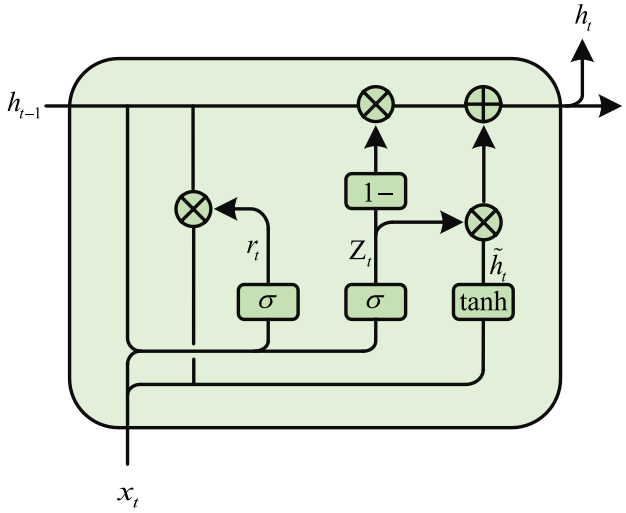


Fig. 7. GRU framework.

$$r_t^l = \sigma\left(W_r^l h_{t-1}^l + U_r^l x_t^l + b_r^l\right) \tag{9}$$

$$z_t^l = \sigma\left(W_z^l h_{t-1}^l + U_z^l x_t^l + b_z^l\right) \tag{10}$$

$$\tilde{h}_t^l = \tanh\left(W_h^l\left(r_t^l \odot h_{t-1}^l\right) + U_h^l x_t^l + b_h^l\right) \tag{11}$$

$$h_t^l = \left(1 - z_t^l\right) \odot h_{t-1}^l + z_t^l \odot \tilde{h}_t^l \tag{12}$$

Here, r_t^l is the reset gate and z_t^l is the update gate, both of which are obtained by applying the sigmoid activation function to a combination of the previous hidden state h_{t-1}^l and the current input x_t^l. The candidate hidden state $\tilde{h}_t^l$ is computed using a tanh activation function based on h_{t-1}^l and x_t^l modulated by the reset gate. The final hidden state h_t^l is a weighted combination of the previous hidden state and the candidate hidden state, where the update gate determines how much of the past information is retained versus how much of the new candidate state is incorporated.

Although GRU improves model efficiency by simplifying the gating mechanism, it still relies on a single hidden state vector to store and update temporal information. As the size of data increases and sequence dependency length extends, GRU may encounter an information bottleneck when modeling long-term dependencies: its limited gating structure struggles to capture extremely long-range contextual relationships, leading to decreased prediction accuracy in complex, large-scale sequence tasks. To address these issues, Zhou et al. (2022) proposed a memory-augmented GRU, which reuses historical state information to slow the forgetting of important trend features. This approach reduced prediction error by 50 % under small-sample conditions compared to standard GRU. He et al. (2024). introduced a model that integrates residual multi-head attention mechanisms with automated feature fusion, enabling joint modeling in the time domain, frequency domain, and time–frequency domain, thereby significantly enhancing the extraction of complex degradation features. These improvements not only enhanced the generalization ability of GRU-based models but also provided more reliable support for real-time maintenance decision-making. To comprehensively compare the performance differences and respective advantages of RNN and its variants in bearing RUL prediction tasks, a comparative analysis was conducted on existing studies in terms of dataset selection, network architecture design, and prediction performance. The results of this analysis are presented in Table 2.

As shown in Table 2, LSTM/GRU performs well in time series modeling with the gating mechanism, but it is still prone to information attenuation and error accumulation in long-range dependence scenarios. Aiming at this defect, a learnable local dynamic memory pool can be introduced into the backbone network to capture short-term high-frequency degradation features. A global graph attention layer is superimposed to map loads, working conditions and multi-source sensor channels into graph nodes, which compensates long-range dependence modeling ability from topology adaptation and mines cross-domain common features. The two-level structure shares the hidden state representation at the same time step, in the early stage with high uncertainty, it focuses on layer inference to bridge the timing blind area, and switches to lightweight memory pool inference to meet real-time requirements after the uncertainty is reduced. In view of the limitations of the existing public datasets in terms of the coverage and distribution diversity of working conditions, the migration and robustness of the framework in complex industrial sites still need further empirical research.

Table 2
Horizontal comparison table of RNN models.

Baseline	Method	Point of improvement	Dataset	Comparative advantage
RNN	RCNN(Wang et al., 2020b)	Cyclic convolutional layers are combined with gating mechanisms to construct a probabilistic prediction framework, enabling the quantification of uncertainty in the prediction process.	FEMTO-ST	Compared to RNN and its variants, CRA metrics achieve optimal performance.
	Encoder-Decoder-RNN(Chen et al., 2020)	Fully automated feature extraction based on frequency band energy eliminates the need for manual construction of health indicators.	PHM 2012	The introduction of attention weight visualization enables intuitive analysis of the features attended to by the model, thereby enhancing interpretability.
	PMCRCNN(Zhou et al., 2021)	Multi-channel convolution is used to extract time-domain, frequency-domain, and time–frequency domain features, integrating global and local features to model the dependencies of degradation characteristics.	XJTU-SY	The multi-channel architecture improves feature representation capability by 22 % compared to single-channel RNNs, achieving RMSE in the range of 0.052–0.094 and MAE in the range of 0.043–0.106.
LSTM	2D-LSTM(Li et al., 2022)	2D-LSTM captures both temporal and spatial dependencies simultaneously and incorporates an IFU designed to enhance the flow of information across network branches.	XJTU-SY, PHM 2012	Compared to 1D-LSTM and Cap-LSTM, it achieves the highest prediction accuracy.
	LSTM-TDNN (Zhang et al., 2023)	LSTM is combined with a nonlinear health indicator function to guide the training process, enabling the learning of degradation trends and RUL prediction through a time-delay network.	XJTU-SY	In the ablation experiments, the RMSE fluctuated around 0.02, with the maximum value remaining below 0.05.
	TSA-LSTM(Li and Jian, 2024)	Combining global search with the temporal sensitivity modeling capability of the LSTM, features are constructed based on the sideband energy ratio.	Measured data of main bearings from offshore wind farms	Compared to LSTM and POS-LSTM, the model achieves RMSE <0.020 and MAPE <0.017.
GRU	AWPME-GRU (Chen et al., 2021)	The Wiener process with adaptive drift rate is combined with GRU to enable accurate prediction of long-term degradation trends.	PHM 2012	A real-time prediction framework is constructed, achieving a CRA of 0.790 and a TMSE of 1.221.
	CMGRU(Zhou et al., 2023)	A unified failure threshold of 0.95 is set using DCRHI, combined with GRU to slow down the forgetting of important trend information.	XJTU-SY, PHM 2012	CMGRU outperforms GDAU, JANET, GRU, and LSTM in both training efficiency and prediction accuracy.
	Attention-MGRU (Shao et al., 2024)	By integrating vibration, temperature, and torque signals, the model employs an attention mechanism to capture the correlations among multi-source features.	Self-collected full-lifecycle data of rotary bearings	The model construction is divided into pre-training and fine-tuning stages. After incorporating fine-tuning, the RMSE is reduced to 0.0442.

4.2. TCN

Although RNN-based models have been widely applied in temporal modeling, their sequential dependency structure limits training efficiency and parallel processing capabilities. TCN, as fully convolutional architectures, effectively alleviate these limitations. The framework of TCN was illustrated in Fig. 8.

TCNs are designed to provide an efficient and easily trainable alternative to RNNs and their variants for sequence-to-sequence prediction tasks (Wang et al., 2021). Unlike RNNs, which perform recursive computations step-by-step along the time axis, TCNs apply convolutional operations over the entire time series simultaneously. By utilizing dilated convolutions and deep network structures, TCNs are capable of capturing long-range historical dependencies while maintaining stable gradients. This not only facilitates easier training but also significantly enhances model robustness (Liang et al., 2023; Tian et al., 2023).

Generally, a TCN consists of three core components: causal convolution, dilated convolution, and residual connections (Qiu et al., 2023). Causal convolution ensures that the model strictly adheres to the causal structure along the time dimension, meaning that predictions at any time step depend only on the current and past inputs-not on any future information. This design aligns with the natural logic of sequence generation and guarantees that each output is conditioned solely on preceding data (He et al., 2022; Wang et al., 2022). Dilated convolution extends the model's receptive field without significantly increasing the number of parameters or the complexity of the architecture. This allows the network to effectively capture long-range dependencies required by long sequences (Zhuang et al., 2021). In contrast, standard convolutions

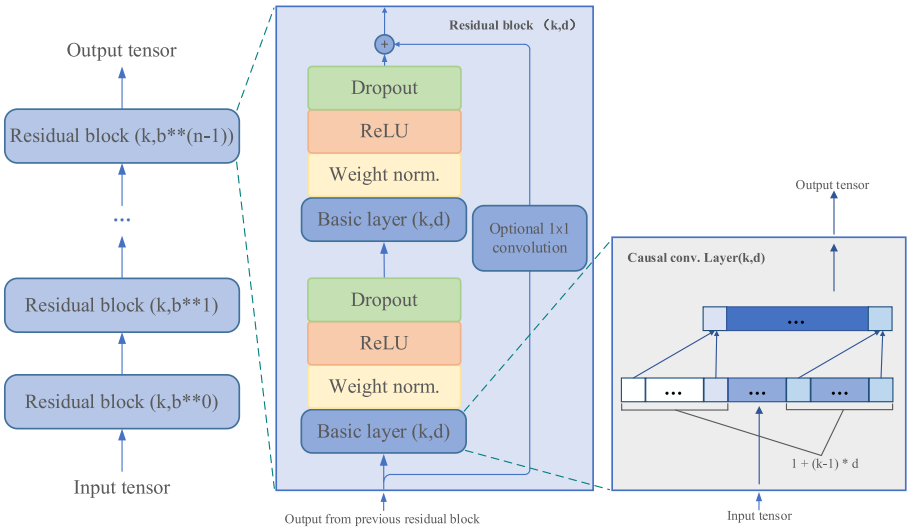


Fig. 8. TCN framework.

require either deeper networks or larger kernel sizes to achieve wide receptive fields (Ma et al., 2024). By inserting dilation factors between filter elements, dilated convolution rapidly expands the receptive field, making it possible to model long-distance dependencies in sequences more efficiently (Yao et al., 2024a).Residual connections ensure effective gradient propagation through deep networks, mitigating the risk of information degradation. They allow the network to focus on learning the residuals (i.e., the differences between input and output), rather than the entire transformation, which reduces training difficulty and prevents issues such as vanishing or exploding gradients (Qin et al., 2024).

The dilated causal convolution layer is shown in Fig. 8. Fig. 8 illustrates a schematic of a regular causal convolutional network. For an input tensor $x_i = \{x_1, x_2, x_3, \ldots, x_N\}, x_i \in R^{T\times N}$ and filter f: $\{1, 2, \ldots, k\} \rightarrow R$,the dilated convolution operation F on the element s of the sequence is defined as Eq. (13)(Bai et al., 2018):

$$F(s) = (x^{*}f)(s) = \sum_{i=0}^{k-1} f(i)\cdot x_{s-d\cdot i} \tag{13}$$

where * denotes the convolution operator, d is the dilation factor, k is the filter size. Specifically, $f(i)$ denotes the i-th element within the filter, $x_{s-d\cdot i}$ denotes the $(s-d\cdot i)$-th element within the input sequence,and $s-d\cdot i$ also accounts for the direction of the past.

Additionally, the issue of gradient vanishing can be effectively mitigated through the incorporation of residual blocks. Given the input x and output $F(x)$ of the layers, the final output of the residual block is represented as Eq. (14):

$$output = Activation(x + F(x)) \tag{14}$$

Where $Activation(\cdot)$ is the activation function, Common activation functions are ReLU, Sigmoid, Tanh, Softmax.

To further enhance the prediction accuracy of TCNs, recent improvement strategies primarily focus on augmenting the traditional TCN architecture with mechanisms such as attention integration, multi-scale feature fusion, and adaptive degradation stage partitioning. One of the major challenges faced by TCNs is the difficulty in fully capturing long-range sequence features, which can lead to the omission of important local information (Cao et al., 2024). Researchers have found that incorporating attention mechanisms into TCNs can effectively address this issue by enhancing the model's ability to retain and emphasize crucial local features. Zhong et al. (2024) proposed a method named TCNAAM, which integrates an adaptive activation function and attention mechanism into the TCN framework. This model adopts a dual-branch architecture: one branch uses an adaptive convolutional scale attention layer to extract key local information, while the other introduces a novel activation function called ULT within the TCN model to suppress noise and retain dominant features. Cao et al. (2021) combined TCN with a residual self-attention mechanism, enabling simultaneous processing of time-domain and frequency-domain information. The RSA mechanism evaluates the contribution of features at each time step, assigning different importance weights to features at different time points, thereby enhancing the model's focus on critical information in the time series. Wang et al. (2023) introduced a Competitive Temporal Convolutional Network that incorporates a Dual Competitive Attention mechanism. A multi-dimensional competitive module is designed, using global max-pooled features as the basis for attention calculation. The attention is applied sequentially along the temporal and channel dimensions, improving the model's feature selection capability. This approach prevents mutual cancellation of informative features during multi-dimensional fusion and ensures the retention of key features.

Multi-scale feature fusion aims to enhance the model's understanding of data by integrating information from different scales. This is particularly effective when dealing with complex and hierarchical features, as it not only helps to eliminate redundant noise and improve model robustness, but also strengthens the model's ability to capture dependencies across long-range information (Zhu et al., 2024). Adaptive degradation stage partitioning ensures that the data within each stage share similar characteristics, allowing the model to focus on critical features specific to each degradation phase (Deng et al., 2022a).Li et al. (2024) proposed an RUL prediction method based on signal reconstruction, where the Teager-Kaiser Energy Operator and VMD are used to reconstruct bearing vibration signals. Teager-Kaiser Energy Operator is employed to extract high-frequency impact energy, while VMD is used to optimize the decomposition process by removing high-frequency noise and effectively capturing weak impact features in the signal. To address instability caused by environmental noise, Wang et al. (2022) introduced an Adaptive Shrinkage Processing method into the TCN framework. This approach applies an adaptive soft-thresholding function to remove uncertain noise from vibration signals, enabling the model to dynamically adjust its signal processing strategy and maintain high prediction accuracy and stability under complex operating conditions.

In addition, some researchers have found that a dynamically plastic temporal receptive field can be achieved without increasing the depth of the network. Risso et al. (2022)designed Lightweight NAS for TCN to achieve parameter compression without accuracy loss, and verified the value of searchable dilation for edge deployment. Lu et al. (2025) used a dynamically learnable dilation factor to automatically adjust the length of the receptive field and significantly improved the accuracy of the C-MAPSS dataset. Based on the TCN model, various improvement strategies and their performance across different datasets are systematically reviewed. The results of this analysis are presented in Table 3.

Based on the research context in Table 3, we can explore a method based on Bayesian or evidence theory to quantify the uncertainty of different degradation stages, and dynamically allocate computing resources and attention weights accordingly. In the early high uncertainty stage, the model automatically shrink the receptive field and strengthen the fine-grained attention to capture weak anomalies. As the degradation features become more prominent and the confidence of the model increases, the receptive field is gradually expanded and the multi-modal long-term dependence is fused to realize the continuous switch from local fine detection to global insight. The framework unifies the three mechanisms of attention enhancement, multi-modal fusion and stage division, and allows the complexity of each mechanism to be weighed according to the task requirements, which promotes the overall rather than incremental performance improvement of the model.

4.3. Transformer

However, with the increasing complexity of bearing degradation processes and the growing need for modeling features across multiple temporal scales, the TCN architecture still exhibits limitations in capturing long-range global dependencies. Against this backdrop, the Transformer architecture has emerged as a promising solution for RUL prediction. Leveraging its self-attention mechanism, Transformer enables global information modeling without relying on a fixed receptive field. This characteristic provides a more powerful and flexible framework for handling complex temporal dependencies.

Transformer was first introduced by Vaswani et al.as a sequence-to-sequence model architecture for natural language processing and other sequential tasks. Its structure is illustrated in Fig. 9.

The SA mechanism within the Transformer architecture enables the model to simultaneously consider information from all positions in the input sequence, effectively addressing the local receptive field limitations inherent in RNN and CNN models that require step-by-step processing (Ding et al., 2023). The SA mechanism allows the model to assign different attention weights to important and less relevant features within the input sequence, thereby facilitating the capture of long-range contextual dependencies (Deng et al., 2022b). The mathematical formulation is expressed as follows:

Table 3
Comparison table of improved models based on TCN model.

Improvement direction	Method	Point of improvement	Dataset	Comparative advantage
Enhancement of the Attention Mechanism	TCN-RSA(Cao et al., 2021)	A residual self-attention mechanism is introduced and combined with TCN to extract features from both the time and frequency domains.	XJTU-SY, PHM 2012	An ablation study was conducted to demonstrate that TCN-RSA outperforms existing methods in both prediction accuracy and computational efficiency.
	Competitive TCN (Wang et al., 2023)	Dual competitive attention, global attention mechanism, and multi-dimensional attention modules are employed.	XJTU-SY, PHM 2012	Through the competitive attention mechanism, RMSE and MAE are reduced by 8.76 % and 8.46 %, respectively, outperforming the traditional TCN.
	TCNAAM(Zhong et al., 2024)	Adaptive activation functions, attention mechanisms, and a dual-branch structure are introduced to locally extract key features.	XJTU-SY, PHM 2012	Compared to FBRA, CAM, DAR, and SA, it achieves the best performance. Relative to models without attention mechanisms, RMSE is reduced by 8.76 % and MAE by 8.46 %.
Multimodal data fusion	Sorting feature-based TCN(Sun et al., 2021)	Combining Local Mean Decomposition with rank-based feature construction enhances the temporal modeling capability.	Dataset from the National Key Laboratory of Mechanical Transmission, Chongqing University	Compared to traditional combinations of EMD and LMD with DL-based time series forecasting models, the proposed approach achieves superior average scores across all evaluation metrics, with a Score of 0.922.
	TKEO-SSA-VMD-TCN(Li et al., 2024)	VMD and TKEO are used to reconstruct the signals, and a dual-channel architecture combining TCN and CNN is employed.	XJTU-SY	Compared to traditional EMD and MCKD methods, it significantly improves the signal-to-noise ratio. The model achieves an RMSE of 0.031 and an MAE of 0.024.
	TCN-BiLSTM (Guo et al., 2024d)	MHA is integrated with TCN and bidirectional LSTM to fuse global degradation features.	XJTU-SY, PHM 2012	3D feature visualization and multimodal fusion significantly enhance generalization capability, reducing RMSE by more than 15 %.
Adaptive phase division	ASP-TCN	An adaptive shrinkage processing module is integrated with TCN to enhance noise suppression capability.	XJTU-SY, PHM 2012	Compared to models such as DSCN and CNN-BiGRU, the average MAE and RMSE are reduced by 52 % and 64 %, respectively.
	Adaptive-Feature-Utilization-SGGTCN (Xia et al., 2023)	A separated gating mechanism is designed to dynamically assess feature importance and model global temporal information.	C-MAPSS, PHM 2012	Adaptive feature utilization combined with multi-kernel maximum mean discrepancy reduces prediction error by 10 %–15 % under compound operating conditions.
	MP-TCN-Mixer (Yao et al., 2024b)	Multi-scale spatiotemporal semantic information is extracted based on a block-wise TCN, combined with an adaptive PAS to eliminate redundant information.	XJTU-SY, PHM 2012	Block-level feature extraction and multi-scale fusion improve RMSE, MAE, and Score metrics by 18.7 %, 16.9 %, and 17.4 %, respectively.

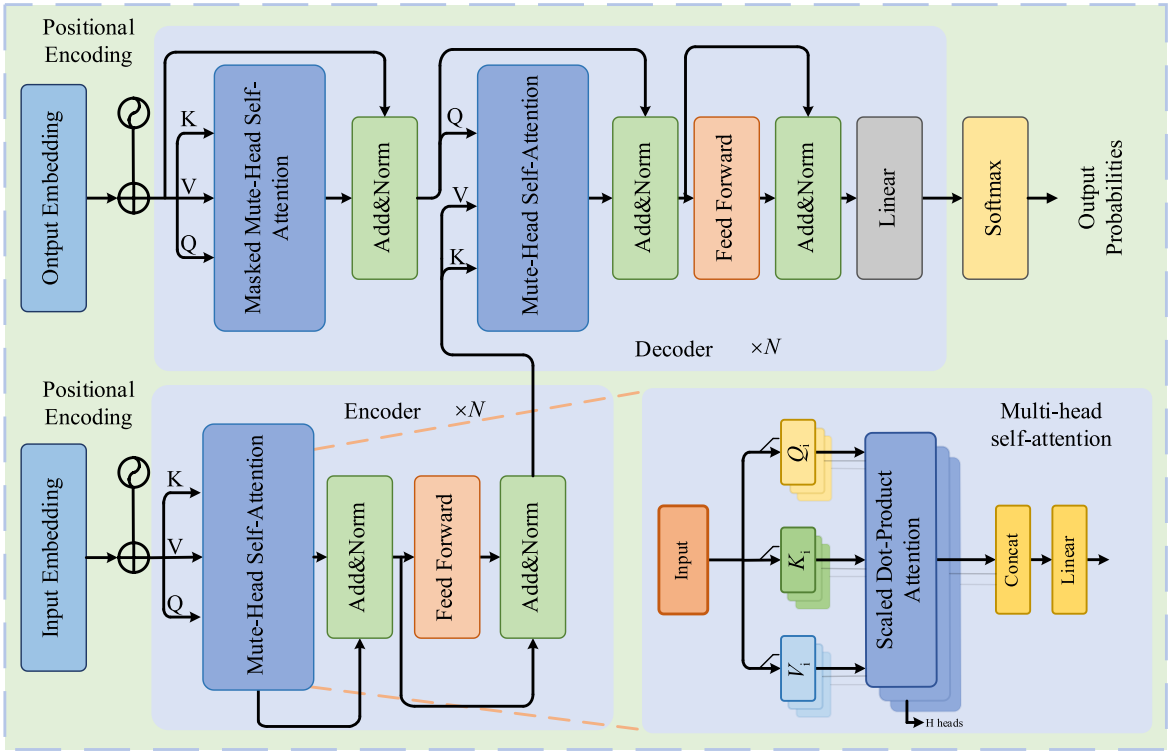


Fig. 9. Transformer framework.

$$Attention(Q,K,V) = soft\max\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V \tag{15}$$

$$soft\max(x_i) = \frac{\exp(x_i)}{\sum_j \exp(x_i)} \tag{16}$$

$$h_i = Attention(Q_i, K_i, V_i) \tag{17}$$

$$MA(Q,K,V) = Concat(h_1, h_2, \ldots h_h)W^o \tag{18}$$

Here, Q, K and V represent the query, key, and value matrices, respectively; d denotes the dimensionality of each attention head. The SoftMax function is applied to perform normalized weighting. The resulting vector h_i is the output of each attention head. Finally, the outputs from all attention heads are concatenated and multiplied by a transformation matrix W^o.

Transformer introduced a fully attention-based sequence transduction model that replaces the recurrent layers commonly used in encoder–decoder architectures with multi-head self-attention mechanisms. However, when directly applied to bearing RUL prediction, the standard Transformer architecture often fails to deliver satisfactory performance. To address this, researchers have combined the strengths of Transformer with other deep learning models to enhance its applicability. Ding et al. (Ding and Jia, 2022) integrated Transformer with a multi-scale convolutional module and sparse self-attention to simultaneously capture global information and local degradation features from vibration signal sequences. This design enables the network to learn rich and shared representations from raw data, significantly improving both prediction accuracy and robustness. Cui et al. (2024b) proposed a Gated Cross-Attention mechanism, which enables adaptive fusion of cross-domain features within a multi-branch architecture. This approach offers a novel perspective for degradation modeling under complex operating conditions.

To address the problem of data distribution shift in real-world industrial scenarios, cross-domain Transformer networks have been proposed. These models leverage dynamic simulation data generation and knowledge transfer to significantly alleviate the challenge of insufficient real-world data (Kim et al., 2024). Zhang et al. (2023) aligned the feature spaces of generated and real-world data, enabling high-accuracy prediction even with only a small amount of measured data. Hu et al. (2024) designed a lightweight Transformer architecture that combines depth wise separable convolutions with MobileViT. This design substantially reduces computational overhead while maintaining high prediction accuracy. These methods collectively demonstrate the remarkable potential of Transformer models in resource-constrained industrial environments. To comprehensively showcase the evolving strategies and effectiveness of Transformer-based models in bearing RUL prediction tasks, a systematic comparison of Transformer improvements is conducted. The detailed results are presented in Table 4.

Combining the three main lines of improvement summarized in Table 4 -structure optimization, technology fusion and optimization mechanism -we can further propose the dynamic sparse Transformer induced by physical constraints. The framework integrates differentiable prior maps derived from physical degradation mechanisms such as power spectral density evolution and stress-strain curve into the attention matrix, implements hierarchical sparsity of attention weights, and replaces the local or frequency domain reinforcement strategy of models such as ConvLSTM-Transformer and CDCT that rely on convolution branches with a topological structure with physical interpretation. The training phase performs gated pruning on the token-level importance based on Bayesian uncertainty to achieve learnable sparse gating. Compared with Global-Local Integration Network and PMBCT, which reduce the calculation amount by decoupling or multi-scale residual, PDST further reduces the attention calculation overhead while retaining high interpretability, and provides a unified physical prior constraint for cross-condition migration.

4.4. Analysis of the applicability of DL-based time series forecasting models

To further explore the advantages and application scenarios of DL-based time series forecasting models in bearing RUL prediction, this study conducts a comprehensive analysis of the applicability of various DL-based time series forecasting models, with the results presented in Table 5.

From the historical context of temporal modeling paradigms, RNN, LSTM, GRU, TCN, and even Transformer form a clear evolution trajectory from "local recursive memory" to "global dependent adaptation". RNN captures short-term fluctuations through explicit state recursion, but gradient attenuation limits its depiction of long-term degradation processes. On this basis, LSTM introduced a gating mechanism to adjust the information flow with a learnable threshold, which significantly extended the memorability interval. GRU further computes the gated structure and reduces the computational overhead while maintaining the long-range memory ability. Then, TCN decomposed the long sequence into parallel convolutional segments by using dilated convolution and residual connection to realize efficient modeling of multi-scale degradation patterns. Finally, Transformer abandons the fixed adjacency topology through multi-head attention, thoroughly decouples temporal dependencies with global relevance weighting, and makes the information flow completely determined by the data itself.

The early model could only resolve point-to-point amplitude fluctuations, but the addition of the gating structure allowed the model to distinguish the key inflection points in the life cycle. The convolutional residual network superimposed and reorganized the degradation features under different time scales, which significantly improved the analytical ability of complex degradation curves. Multi-head attention further integrates multi-source information, realizes cross-modal and cross-condition invariant representation, and lays a foundation for condition transfer in complex industrial scenarios. This is followed by a jump in prediction accuracy as well as higher interpretability of the deep degradation mechanisms.

However, the improvement of model capability is accompanied by the parabolic growth of resource consumption and inference delay. In edge deployment scenarios, the model with the best performance is not necessarily the model with the best value. The practical choice has shifted from "pursuing absolute accuracy" to the multi-objective compromise of "maximizing residual value under cost constraints", which has given birth to lightweight research such as sparsity, quantification and structure reuse, and also prompted the academic community to consider model design, operation and maintenance strategy, and hardware ecology together.

5. Hybrid models: combining prediction steps for enhanced RUL performance

5.1. Solve noise interference and data quality deficiency

Noise interference and insufficient data quality from experimental acquisition are common issues in bearing RUL prediction tasks (Deng et al., 2024). High-frequency noise during vibration signal acquisition, electromagnetic interference in industrial environments, and noise caused by mechanical vibration transmission can introduce high-frequency components that mislead the model into learning spurious features, resulting in overfitting. Moreover, discrepancies in feature distributions between training and testing data significantly reduce the model's generalization capability (Song et al., 2023). Additionally, full-lifecycle data from industrial equipment is difficult to obtain, and data collected under different operating conditions is often imbalanced. Inaccurate RUL labels can cause the model to learn incorrect dependencies, thereby increasing the risk of underfitting (Yang et al., 2024b).

A single model often struggles to effectively mitigate noise

Table 4
Comparison table of improved models based on Transformer model.

Improvement direction	Method	Point of improvement	Dataset	Comparative advantage
Model-based architecture improvements	ConvLSTM-Transformer (Deng et al., 2022b)	Spatiotemporal features are extracted using convolutional LSTM, and long-term degradation dependencies are modeled with Transformer.	XJTU-SY, PHM2012	The advantage of long-term dependency modeling is validated, achieving an RMSE of 0.023.
	CDCT(Jiang et al., 2023)	Time-domain and frequency-domain signals are fed into a dual-channel parallel structure, where causal convolution is used to enhance local features, followed by a self-attention mechanism to strengthen feature weighting.	XJTU-SY, PHM2012	Compared to CBLSTM, DARNN, and TCN_RSCB, the proposed method yields the smallest area in the radar chart across MAE, RMSE, and Score metrics.
	cross-transformer fusion segmented data cleaning (Hou et al., 2024)	Noise penalty factors are assigned separately across different degradation stages, enabling cross-stage feature fusion.	PHM 2012	RMSE and MAE are improved by 1.88 and 5.78, respectively, with the proposed method achieving a 59.10 % improvement in RMSE.
	WTE-Trans(Zhang et al., 2024)	A weighted temporal embedding module is combined with 3D convolution to fuse multi-scale temporal features.	PHM 2012	The proposed method achieves the lowest RMSE compared to all other embedding approaches.
	LECformer(Zhang et al., 2025)	Locally enhanced channel self-attention is employed to dynamically weight features within multi-sensor channels.	C-MAPSS, PHM2012	The average RMSE is reduced by 41.7 %, MAE by 33.4 %, and the Score metric is improved by 22.2 %.
Improvement in the direction of technology application	SAConvFormer(Chen et al., 2022)	Spatial attention enhances convolutional features, which are combined with Transformer for global modeling, eliminating the need for manual feature engineering.	XJTU-SY, PHM 2012	Compared to the baseline model SAConvFormer, the algorithm performance is reduced by approximately 0.08 in RMSE.
	Self-Supervised Transformer (Lin et al., 2024)	The model architecture, after pretraining with CPC and subsequent fine-tuning, utilizes partial lifecycle data instead of full-lifecycle data.	XJTU-SY, PHM 2012, CWRU	The proposed method achieves RMSE values below 0.1 on 19 datasets, whereas methods based on BiLSTM and MSCNN mostly yield RMSE values exceeding 0.1.
	Gate-CNN and Conv-transformer encoder (Huang et al., 2024)	Gated convolution is used to suppress high-frequency noise, followed by the integration of a multi-head attention mechanism to enhance effective degradation signals.	UT6818, PHM2012	The average MAE is reduced by 26 %, the average RMSE is reduced by 22 %, the average R^2 increases by 3 %, and the accuracy increases by 8 %. The average standard deviation decreases by 21 %.
	DCTN-CBAM(Guo et al., 2024b)	The model is extended to a Bayesian deep neural network through variational inference, resulting in more prominent representation of features within the corresponding confidence intervals.	XJTU-SY	The lowest RMSE achieved is 0.111, and the lowest MAE is 0.058.
The improvement is based on the model's own optimization direction	Global and Local information integration network(Chen et al., 2023)	The designed cross-attention discrepancy uses prior knowledge of consistent positions to correctly align similar degradation features.	CMAPSS, PHM 2012	Under complex operating conditions and fault modes, the proposed method achieves an average score reduction of 5.9 % compared to the state-of-the-art methods.
	PMBCT(Peng et al., 2024)	The multi-scale Bayesian convolutional self-attention layer quantifies uncertainty and, when combined with knowledge distillation, enhances the model's generalization ability.	PHM2012, Self-measured complete gear degradation dataset.	The PICP reaches as high as 95.15 %.
	ProgSViT(Hu et al., 2024)	Depth wise separable convolutions are used to extract local features, while MobileViT optimizes computational efficiency.	PHM 2012	While maintaining high computational efficiency, the accuracy is on par with other advanced models.

interference and cannot fundamentally overcome the limitations posed by poor data quality. To address this, researchers have adopted hybrid modeling approaches that leverage the strengths and advantages of different methods or architectures to specifically target noise-related challenges. For example, Zuo et al. (Tao Zuo et al., 2023) proposed the HA-ConvLSTM hybrid model, as illustrated in Fig. 10. In the data preprocessing stage, multi-wavelet decomposition is applied to the raw signal using wavelet packet decomposition to initially suppress the

Table 5
Comparison of the applicability of DL-based time series forecasting models.

Model	Key features	Applicable scenarios	RUL prediction advantage	Ref.
RNN	The basic recurrent structure handles sequence data.	Short sequences and low-complexity tasks.	Short-term vibration signal analysis captures the local fluctuations of the vibration signal, making it suitable for quick deployment.	(Wang et al., 2020b; Zhou et al., 2021)
LSTM	The introduction of gating mechanisms helps mitigate the issues of gradient vanishing or explosion.	Long sequences, capturing the full lifecycle degradation process.	The gating mechanism filters noise, and the input gate adaptively adjusts the feature weights of the bearing under different load conditions.	(Li and Jian, 2024; Ma and Mao, 2021; Wang et al., 2022)
GRU	Optimized gating mechanism in LSTM.	Resource-constrained scenarios.	Fast inference, supporting online health monitoring.	Zhou et al. (2022)
TCN	Combining causal convolution, dilated convolution, and residual connections.	Well-suited for capturing features across multiple time scales.	Multi-stage, multi-scale degradation feature analysis, with parallel computation for industrial big data processing.	(Song et al., 2021; Wang et al., 2021)
Transformer	Multi-head attention and positional encoding.	Long-range dependencies across time steps.	Captures features from multi-source data, with strong cross-condition generalization capability.	(Guo et al., 2024a; Li et al., 2024; Su et al., 2021)

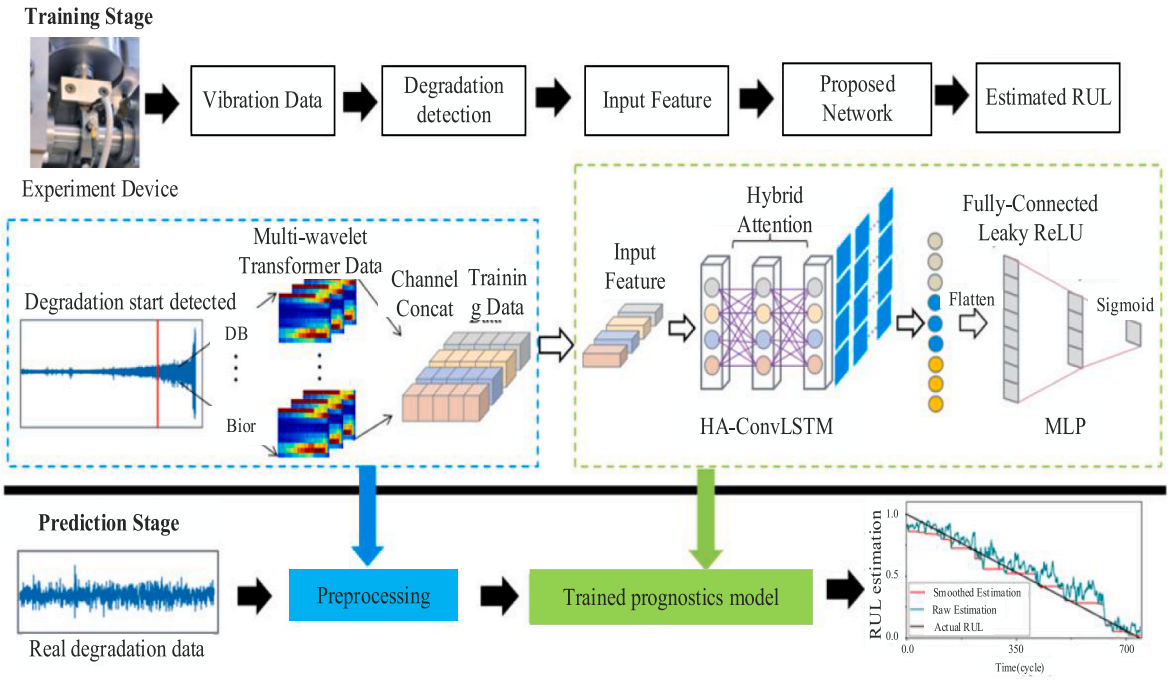


Fig. 10. HA-ConvLSTM (Tao Zuo et al., 2023).

impact of noise. The frequency band coefficients obtained from different wavelet bases are then concatenated into a two-dimensional time–frequency image, which serves as the model input. This enhances the comprehensiveness of signal representation and reduces the influence of noise concentrated in any single frequency band. A channel attention mechanism is inserted between ConvLSTM layers, enabling the model to focus on meaningful degradation features even in the presence of noise. This approach effectively combines multi-wavelet coefficient fusion and attention mechanisms to suppress noise, while leveraging ConvLSTM for long-range dependency modeling.

Song et al. (2023) proposed the MSDRSN model, in which a Deep Residual Shrinkage Network is employed with a soft-thresholding denoising mechanism to adaptively filter out noise. Multi-scale convolutional kernels are used as base learners to capture both local and global features across different scales, thereby reducing the sensitivity of the model to noise in any single scale. Wen et al. (2024) introduced the Loop-GAN-GRU-AE-Wiener approach, as illustrated in Fig. 11. This method uses Loop-GAN to generate synthetic data, addressing the issue of insufficient training samples. The generated data is further enhanced through wavelet transform and cumulative transformation, which improve signal trend representation while eliminating redundant noise. The GRU is integrated with a Wiener process, which incorporates prior knowledge from physical models to strengthen the model's robustness.

5.2. Solving the lack of feature relationship modeling

In bearing RUL prediction, feature selection and the relationships among features play a crucial role in determining the model's prediction accuracy (Yao et al., 2024a). Failure to capture key features or the nonlinear interactions between them can hinder the model's ability to learn effectively, increase computational complexity, and slow down convergence (Wang et al., 2020). This is especially true when dealing with long sequence data, such as full-lifecycle bearing vibration signals. If the temporal and spatial dependencies among features are neglected, the model may fail to grasp the long-term trends embedded in the vibration signals, resulting in inaccurate RUL predictions (Dong et al., 2023). Therefore, effective feature extraction and selection, along with proper modeling of temporal and spatial dependencies, are critical factors in improving prediction performance.

To address the challenge of inadequate feature relationship modeling, researchers have proposed a variety of improvements, mainly in three directions:

Attention-based modeling methods aim to automatically capture intrinsic relationships among features and assign different weights to different features. This enhances the model's sensitivity to key feature combinations, thereby improving prediction accuracy. By emphasizing important features and suppressing redundant or irrelevant information, attention mechanisms also enhance the model's generalization capability (Chang et al., 2022).

Deep feature fusion methods integrate features from multiple sources or scales to overcome the limitations of single-feature inputs in modeling complex nonlinear relationships. By constructing multi-level feature representations, these methods extract cross-scale and cross-channel information, improving the model's ability to represent the evolution

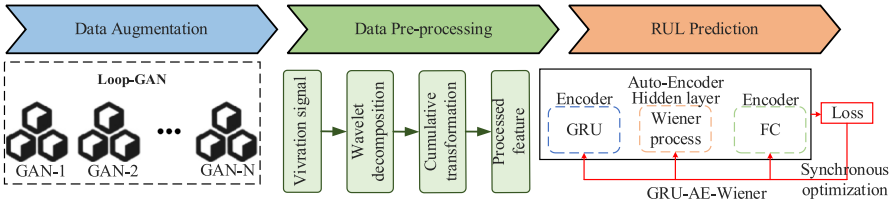


Fig. 11. Loop-GAN-GRU-AE-Wiener(Wen et al., 2024).

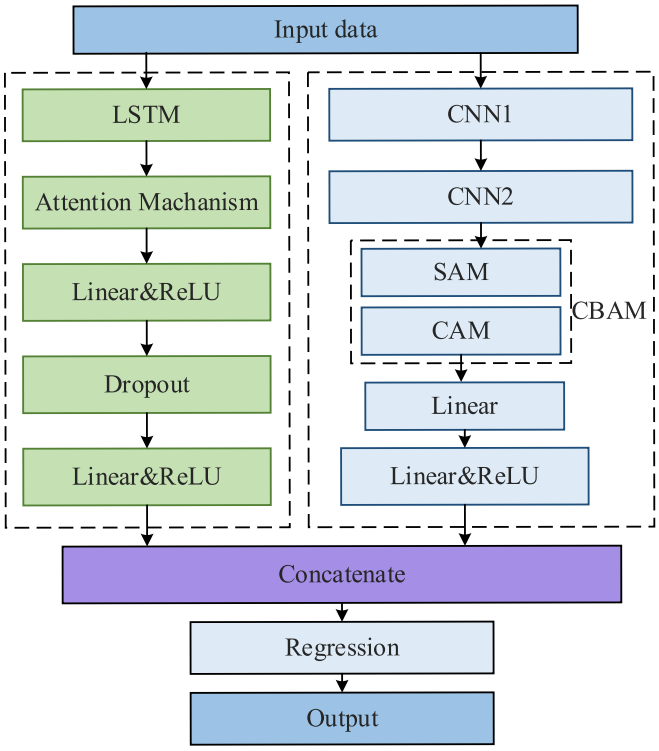


Fig. 12. CALAP(Wu et al., 2023).

of bearing health states and thereby enhancing prediction accuracy (Wang et al., 2024).

Transfer learning and domain adaptation methods are primarily used to address issues arising from varying operating conditions or mismatched data distributions between training and testing. By introducing domain adaptation techniques, these methods adjust the feature mappings across different tasks or data domains, reducing the negative impact of distribution shifts and enabling the model to maintain high prediction performance under diverse operating conditions (Xia et al., 2024).Wu et al. (2023) proposed the CALAP hybrid model, as illustrated in Fig. 12.

This model incorporates attention mechanisms into two parallel feature extraction paths: CNN and LSTM. It addresses the problems of improper feature weight allocation and the loss of long-term dependencies. CNN excels at extracting spatial relational features but is less effective for long time-series data and often under-emphasizes important features. LSTM, while good at modeling temporal dependencies, is prone to forgetting long-term information. By strategically embedding attention mechanisms in both parallel channels, CALAP combines the strengths of CNN and LSTM, significantly improving the accuracy and robustness of RUL prediction.

When processing multimodal data, different feature domains often contain complementary information, necessitating a flexible fusion mechanism to enhance prediction performance. To this end, Su et al. (2020) proposed the MS-DRNN hybrid model, which performs parallel processing of time-domain, frequency-domain, and time–frequency domain features through multi-stream data input. A fusion layer is introduced at the end of the Deep Recurrent Neural Network to jointly model key information from different feature domains. Notably, this fusion strategy can dynamically select the most suitable fusion approach based on the characteristics of each specific data point. To address the challenge of distribution discrepancies across different bearings, the TMLAN hybrid model (Dong et al., 2023), shown in Fig. 13, was proposed.

This model inputs both source domain and target domain data into the TMLAN network for joint training, enabling the extraction of domain-invariant features and facilitating knowledge transfer from the source to the target domain. A convolutional autoencoder is trained to extract deep features from bearing data, and the Pearson correlation coefficient is computed to construct a health indicator. A Bi-LSTM

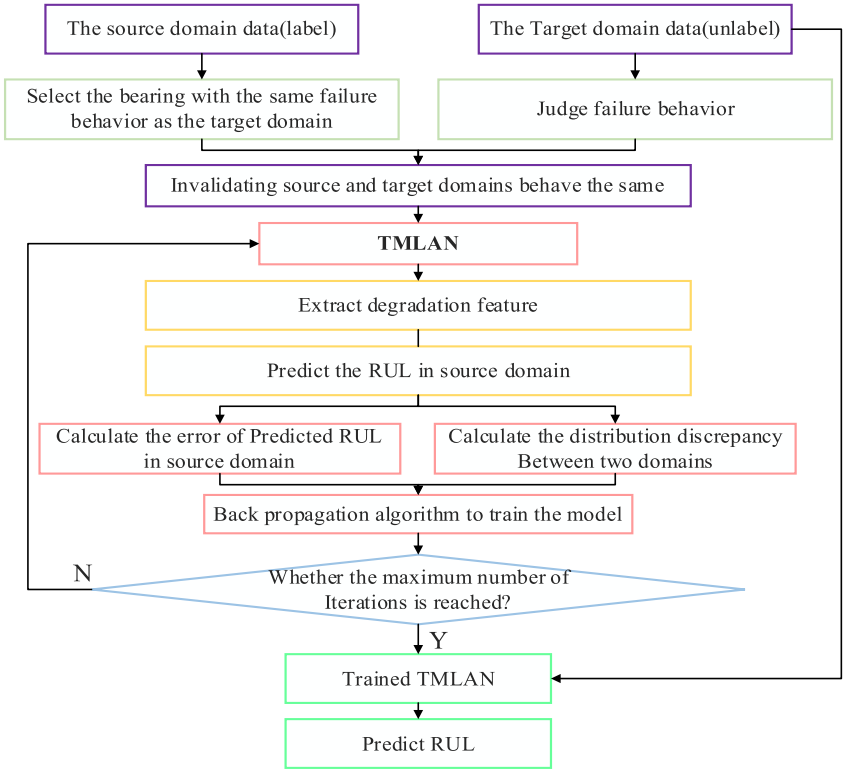


Fig. 13. Deep transfer learning based on Bi-LSTM and attention framework(Dong et al., 2023).

combined with an attention mechanism is used to focus on key features at critical time steps. Domain adaptation is achieved using MMD to align features across different bearings. Compared to models without transfer learning, this approach significantly improves prediction performance, reducing RMSE from 0.25 to 0.08-a 68 % reduction.

5.3. Lightweight design issues

In real-world industrial applications of RUL prediction for bearings, two core challenges persist: lightweight model design and long-range dependency modeling (Abbasi et al., 2024; Zhou et al., 2024). Lightweight design primarily focuses on parameter compression and computational efficiency. Introducing lightweight recurrent neural networks has become one optimization approach, in which redundant gating units are removed to reduce model complexity. This not only decreases the number of model parameters but also improves computational efficiency and enables parallelized training, thereby reducing resource consumption while maintaining predictive performance (Wei et al., 2022). Researchers have also explored the integration of physical models with DL-based time series forecasting models as well as the combination of classical statistical methods and deep learning approaches to improve the accuracy of RUL predictions (Ghatage et al., 2023). For instance, Shi et al. (2024) proposed the DA-LSTM model, which incorporates a Feature Aggregation Attention mechanism into the LSTM decoder to retain key information from raw sensor data. The model also uses ES to focus on the trend components of sensor signals. This design maintains high prediction accuracy while significantly reducing the number of parameters. On the C-MAPSS dataset, DA-LSTM achieved comparable or superior performance to other attention-based hybrid networks with only 3350 parameters and an order of magnitude fewer FLOPs. To address the high computational demand of deep convolutional networks while achieving lightweight design, Wang et al. (2023) developed the BLKAN hybrid model. It employs a Bayesian kernel decomposition strategy, which breaks down large-kernel convolutions into multiple smaller convolution operations, greatly reducing computational overhead. As shown in Fig. 14, the Bayesian large-kernel convolution is decomposed into three simplified components: Bayesian depth wise convolution, Bayesian dilated convolution, and Bayesian pointwise convolution. Compared to models without kernel decomposition, BLKAN reduces computational cost by approximately 58 %.

Yang et al. (2024) introduced the Lightweight Probabilistic Spatio-temporal Network, which significantly reduces the number of model parameters and improves spatial feature extraction efficiency by adopting the SqueezeNet structure in the convolutional layers. For temporal modeling, they proposed a WDRU, as illustrated in Fig. 15.

This unit merges the forget gate and input gate of LSTM and removes the independent output gate, directly computing the hidden state to further reduce computational overhead. Experimental results showed that LPST-Net achieved an average RMSE of 0.29 and an average MAE of 0.17, representing an 81.9 % reduction in error compared to GPR-Net, and a 75.4 % reduction compared to Bi-WDRU-GPR-Net.

5.4. Address model interpretability

DL based-time series models and their hybrid variants have demonstrated outstanding performance in industrial applications such as bearing RUL prediction, equipment health monitoring, and fault diagnosis. However, their inherent "black-box" nature has limited their adoption in critical systems (Cheng et al., 2024). To address this, researchers have proposed various hybrid modeling strategies aimed at

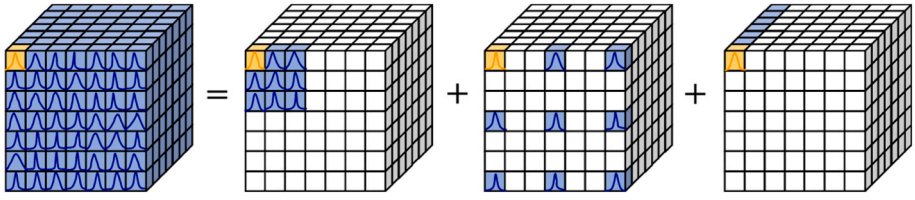


Fig. 14. The kernel decomposition diagram of Bayesian large-kernel convolution (Wang et al., 2023).

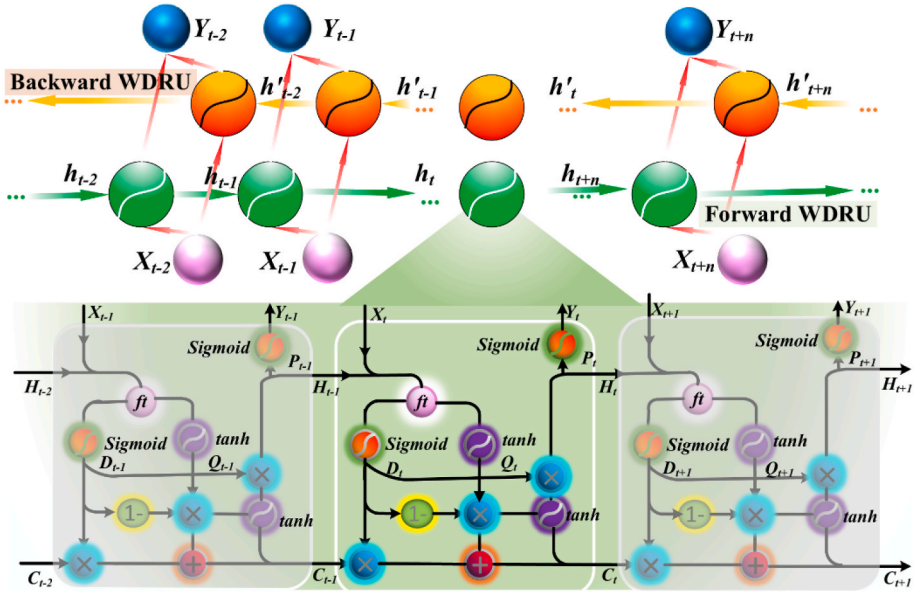


Fig. 15. Structure diagram of bidirectional weight diminishes recurrent unit network(Yang et al., 2024).

enhancing model interpretability. These strategies not only ensure accurate predictions but also provide insights into the model's decision-making processes, thereby significantly improving the transparency and trustworthiness of the models (Liao et al., 2023).

Methods for improving model interpretability generally fall into two categories: Feature importance analysis and degradation process visualization, which aim to reveal the key time steps and core features that influence prediction outcomes. The central idea involves integrating attention mechanisms with visualization techniques. For instance, the cross-attention distribution in Transformer architectures can intuitively illustrate the relationships between monitoring data and degradation states, thereby enhancing the model's intrinsic interpretability (Chen et al., 2023). Some studies have also proposed frameworks such as FFVE, as shown in Fig. 16.

This framework utilizes factorized spatiotemporal and channel fusion blocks to reduce redundancy in sensor data while compressing the data into a three-dimensional latent space, enabling clear visualization of the degradation process (Chen et al., 2024).

Another direction focuses on integrating DL-based time series forecasting models with physical modeling, typically by incorporating physical constraints into the loss function or feature construction stage to ensure that prediction results align with real-world mechanisms. One such approach is the physics-guided degradation trajectory modeling based on phase space reconstruction, which treats RUL prediction as the estimation of changes in phase space trajectories. As shown in Fig. 17, this method derives degradation paths that better comply with physical laws by leveraging physics-informed inputs (Yin et al., 2025). Additionally, some researchers have proposed incorporating physics-guided inconsistency loss functions into stacked Bi-LSTM networks, forming a multi-objective loss to ensure that the model's predictions remain consistent with known physical phenomena (Zhu et al., 2024).This guarantees the physical plausibility and reliability of the model outputs.

5.5. Analyze solutions to key problems

In bearing RUL prediction tasks, models deployed in real industrial applications face a range of challenges, including data noise interference, insufficient feature modeling capability, the need for lightweight architectures, and a lack of interpretability. To effectively address these issues, researchers have proposed a variety of hybrid network strategies, focusing on systematic optimizations across key aspects such as data preprocessing, feature representation, structural compression, and decision transparency. The specific strategies and their corresponding optimization outcomes are summarized in Table 6.

In Table 6, the four common pain points in bearing RUL prediction are corresponding to the three strategies of data layer, model layer and decision layer, so as to split the complete workflow into three pluggable links: front-end data processing, mid-end network architecture and

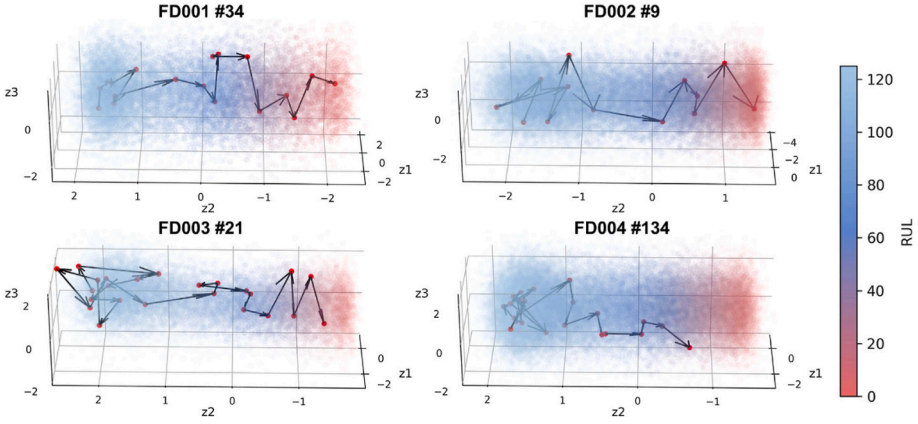


Fig. 16. Degradation path representation of FFVE under four operating conditions (Chen et al., 2024).

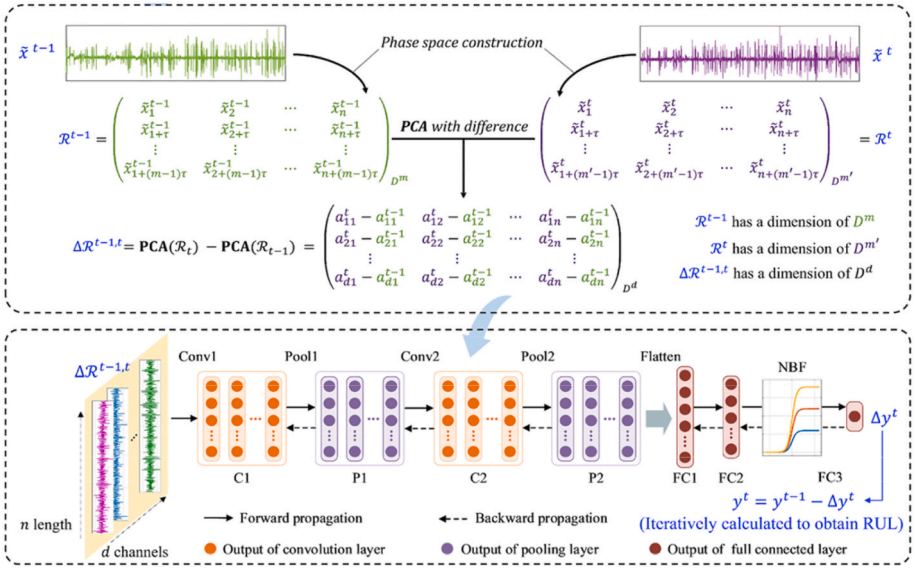


Fig. 17. Physics-guided degradation trajectory model (Yin et al., 2025).

Table 6
Analysis of solutions to key problems in RUL prediction.

Core problem	Data-Centric	Model-Architecture-Centric	Decision-Centric
Noise interference & data scarcity	Wavelet packet decomposition combined with GAN/VAE; Mask combined with self-supervised pre-training (Cen et al., 2023; Z. Chen et al., 2024)	Bayesian with convolutional structure(Feng et al., 2024; Zaiem et al., 2025)	Bayesian confidence decision(Leadbetter et al., 2025)
Insufficient modelling of complex dependencies	Time and frequency domain integration of multiple views; Physical Prior Data augmentation(Kulevome et al., 2024; Russell and Wang, 2022)	Backbone network combined with attention mechanism; Physical residual module is added to trunk network(Li et al., 2025; Pandiyan and T., 2025)	Multi-sensor post fusion(Makrouf et al., 2025; Xu et al., 2024)
Light-weight deployment	Compression of high-dimensional signals	network structure compressed and sparse(Wang et al., 2023; Xiang et al., 2024)	Early-Exit(Bajpai and Hanawal, 2025)
Interpretability	Counterfactual data augmentation(Andringa et al., 2025)	Adaptive Concept Bottleneck Model (AdaCBM, 2024)	Counterfactual decision auditing(Andringa et al., 2025; Cummins et al., 2024)

back-end inference and decision, and perform multi-angle compensation and performance enhancement for the same problem through cross-layer collaboration.

In the face of noise interference and data scarcity, the front-end can use wavelet decomposition combined with generative augmentation or self-supervised mask to improve signal quality and sample diversity at the input. In the middle, Bayesian convolution or other uncertainty layers are introduced to give confidence to the feature representation. The back-end uses Bayesian confidence integration or rejection mechanism to reject the unreliable prediction.

For the lack of complex dependency modeling, the front-end firstly performed multi-view time-frequency fusion or superimposed physical prior augmentation to provide complementary information for the network. In the middle end, convolution and attention are combined or physical residual units are inserted to capture local patterns, global associations and mechanism constraints. The back-end uses multi-sensor post-fusion, which can perform consistent calibration of different source results.

In the lightweight deployment scenario, the front-end firstly compresses high-dimensional signals to reduce data transmission and storage overhead. In the middle end, the number of parameters is reduced by structure pruning, sparsification or searchable lightweight architecture. The backend ends the inference dynamically using an early departure mechanism.

In order to improve the interpretability, the front-end introduces counterfactual data augmentation to generate control samples to help discover key features. At the middle end, explicit semantic layers such as concept bottlenecks were added to expose the middle reasoning process. The backend gives causal paths through counterfactual decision audit attributions.

Table 6 is essentially a combined toolbox of pain points, strategies, and hierarchical mapping. For specific application, engineers can first locate the most prominent problems, and then combine resource constraints and regulatory requirements, select appropriate strategies from the front-end, mid-end, and back-end layers to assemble, and optimize the trade-off between multiple objectives. Finally, they can build a RUL prediction pipeline that is robust, interpretable, and easy to deploy.

6. Existing challenges and future perspectives

6.1. Existing challenges

The current contradictions that restrict the application of bearing RUL to predict depth are distributed in gradient. The first is that it is difficult to quantify the effectiveness of the attention mechanism. Followed by the hybrid model in the interpretability and resource precision tradeoff double bottleneck; Realizing millisecond level online inference and maintaining cross-condition robustness in the case of scarce samples belong to subsequent topics that need to be laid out in advance when further large-scale landing.

(1) The applicability of attention mechanisms is difficult to assess

Attention mechanisms have garnered widespread interest in the academic community due to their ability to automatically assign varying weights to input features, thereby highlighting the relative importance of each feature. However, in real-world industrial scenarios-where higher levels of safety, stability, and interpretability are required-the practical value of attention mechanisms becomes more difficult to assess. The weight distributions produced by attention layers often lack direct correspondence to physical processes or engineering intuition, making the interpretability and effectiveness of these mechanisms difficult to reliably verify (Niu et al., 2021). While attention modules offer noticeable performance gains in randomly initialized and shallow networks, such gains diminish-or even become detrimental-in deep, large-scale pre-trained networks. This suggests that the utility of attention mechanisms must be quantitatively evaluated for specific tasks to ensure their effectiveness. Moreover, attention mechanisms compute pairwise similarities across all tokens in the input sequence, resulting in a quadratic growth in computational complexity relative to sequence length. This makes them resource-intensive, especially for long time series. Additionally, attention may inadvertently assign high weights to noisy or irrelevant features, increasing the risk of overfitting and reducing the clarity of the model's true performance contribution (Hernández and Amigó, 2021). Kim et al. (2020) explored the uncertainty of attention mechanisms by designing and comparing four different attention frameworks. Their results showed that the distribution of attention weights is sensitive to parameter choices, which can introduce bias or overfitting under certain configurations. Furthermore, features receiving high attention weights did not necessarily align with the key features identified by traditional machine learning methods. In scenarios with limited prior knowledge, relying solely on attention-derived "important" features can lead to the misidentification of spurious correlations as key indicators. In order to further clarify the attention mechanism, this paper systematically sorts out the core issues of the reliability evaluation of the attention mechanism in Table 7, and gives the corresponding test schemes and quantitative indicators.

Table 7 shows that the reliability of the attention mechanism primarily depends on whether its weights can be mapped to stress waveforms or characteristic frequencies; Otherwise, even if the evaluation indicators rise, it will still be impossible to obtain engineering recognition. Secondly, the model must maintain consistency under different working conditions. This prerequisite determines whether the algorithm can truly be implemented. Specifically, in terms of task objectives: If early fault detection is the top priority, the recall rate should be given priority consideration. If the focus is on optimizing the maintenance plan, attention should be paid to the smoothness and stability of the RUL curve. Overall, classical physical features provide a global and robust explanatory framework, while the attention mechanism supplements local details on this basis, achieving complementary advantages of the two.

(2) DL-based time series forecasting models still have shortcomings in terms of interpretability

Table 7
Core problems and evaluation framework for attention mechanism reliability.

Core problem	Test scheme	Quantitative metrics
Does the attention weight correspond to physical mechanisms such as stress wave propagation and characteristic frequency?	The RUL error was observed after shielding the high-weight frequency band. Cross correlation tests the correlation between weights and physical quantities.	RUL error increase >15 % and Correlation coefficient >0.6 regarded as established
Is attention still effective when working speed and load change?	Training across working conditions, the consistency of weight distribution under different working conditions was compared.	Kendall $\tau \geq 0.6$
Is performance improved by error detection and recall, or by lifetime curve smoothing?	Multi-objective experiment: Maximizing recall rate and curve smoothness respectively	Recall goes up $\geq$ 5 pp; Curve fluctuation decrease$\geq$ 10 %
Is it to replace or complement classical features such as RMS and kurtosis?	Compare the results of only traditional features, only attention and fusion features	F1Score, RMSE

Even though DL-based time series forecasting models have demonstrated exceptional adaptability and nonlinear modeling capabilities in RUL prediction tasks, their internal decision-making logic often remains inaccessible to engineers (Hassija et al., 2024). The specific analysis of interpretability is shown in Table 8.

As shown in Table 8, only when the explanatory requirements are tightly coupled with the decision scenarios, stakeholders and risk costs, can the model with practical application value be constructed. In this framework, interpretability is the core mechanism to ensure the reliability of decision making, rather than an auxiliary feature.

(3) DL-based time series forecasting models still face important challenges in industrial deployment

In industrial environments, there are often stringent requirements for hardware resources, deployment cost, and model response speed. Edge computing devices and embedded systems typically have limited computational power and storage capacity, necessitating the use of model compression techniques such as pruning and knowledge distillation to achieve lightweight deployment (Gao et al., 2024). In the following, the key points of lightweight deployment are analyzed to achieve real-time, safe, and energy-efficient deployment of RUL prediction, and the analysis is shown in Table 9.

As illustrated in Table 9, lightweight deployment can be structured into four stages: first, determining network scale and tolerance limits through reverse inference based on predefined accuracy thresholds; second, iteratively compressing the model by progressively eliminating redundant parameters; third, strategically allocating parameters to suppress cumulative errors; and finally, selecting optimal trade-offs between power consumption and latency based on Pareto front analysis. Fundamentally, lightweight deployment aims to simultaneously optimize decision credibility and computational resource constraints.

(4) How to realize online real-time prediction is an urgent problem to be solved

In practical industrial operations, bearings require real-time monitoring and rapid feedback to promptly detect faults and initiate effective maintenance actions. Online RUL prediction involves continuous sampling of sensor data, which imposes high demands on the throughput and response speed of the deployed networks (Puerto-Santana et al., 2022). Table 10 discusses the core issues it faces.

Table 11 highlights the system-level balance of online health prediction: on the one hand, the prediction lead and end-to-end timeliness

Table 8
Explanation requirements and evaluation metrics under different decision scenarios.

Decision scenario	Necessary explanation granularity	Recommended explanation methods	Evaluation metrics
Spare Parts Management	Module attribution & major time-frequency visualization	Hierarchical SHAP	explanation score $\geq$4/5
Real-Time Risk Assessment	Counterfactual samples	Minimal counterfactual sequence generation	Counterfactual feasibility >80 %
Algorithm Tuning	parameter sensitivity	hyperparameter-sensitivity heatmap	Top-k feature-gradient consistency $\geq$0.7

Table 9
Key issues, strategies, and target metrics for lightweight deployment.

Key issue	Recommended lightweight strategy	Target metric
compute Platform & power budget	Convert model to TensorRT/TFLite(Howard et al., 2019)	Inference latency $\leq$50 % of real-time control cycle
Acceptable accuracy loss	Structured magnitude pruning(Han et al., n.d.)	ΔRMSE/baseline $\leq$20 %
Precision safeguard for sensitive layers	keep critical layers in FP16, convert others to INT8/INT4(Micikevicius et al., 2018)	Error amplification confined to first and last or attention layers
Energy–performance Pareto Optimization	quantization-aware training(Zhou et al., 2017)	power reduction with <5 % accuracy loss

Table 10
Deployment feasibility optimization for real-time health management.

Key issue	Recommended strategy	Evaluation criteria
Fault Lead Time	Derive the minimum acceptable horizon from the maintenance critical path	Horizon $\geq$1.5 $\times$ average replacement duration
chains increase information staleness	Compress total latency via edge inference + model compression + zero-buffer data pipelines	Total latency $\leq$25 % of the signal's time constant
Sliding window may waste computation	The bottom fixed sliding window ensures the complete data flow. The upper layer triggers incremental inference based on health index	Window width = minimum length required to suppress noise
Point RUL vs. Degradation Trajectory	Output both Time-to-Failure and confidence bounds	Point estimate MAE; quantile loss

Table 11
Domain adaptation and label efficiency strategies under data constraints.

Key issue	Recommended strategy	When to apply
Generalization across machines, materials, and operating conditions	• Define the fleet and operating envelope during planning • Quantify domain gaps with simple distance measures • Learn domain-invariant or meta-adaptable representations	Deployment scenarios where the target equipment set is known in advance
Dual bottleneck of label cost and data-collection time	• Semi-/weak-supervised labeling and active learning • Synthetic or simulation data to enlarge coverage • Online label refinement with confidence checking	Situations with limited budgets or sparse fault samples
Benchmarking transfer difficulty	• Provide two benchmark levels: (i) same machine, new load; (ii) new machine, new load • Release datasets with transfer tags and common metrics	Community or competition benchmark-building stages

jointly determine the reliability of the decision window. On the other hand, the granularity of data management and reasoning affects the computational cost and the fineness of risk characterization. Through the checks and balances of these four dimensions, the framework emphasizes the synchronization of physical degradation and digital reasoning rhythm under the premise of meeting the critical path of operation and maintenance, and supports risk-cost decision-making with multi-scale output.

(5) The problem of insufficient generalization and scarce data sample

The distribution of sensor-collected data can vary significantly due to changes in load, rotational speed, lubrication conditions, or external environmental factors. These variations often lead to a noticeable degradation in model prediction performance when applied to new operating conditions (Ni et al., 2023). In order to solve the problem of insufficient generalization and sample scarcity, the following table discusses the three core issues.

As shown in Table 11, first of all, the applicable models and working condition boundaries of the model must be clearly defined in the planning stage; otherwise, any migration discussions will lose their basis. Secondly, the cost and cycle of data and label acquisition determine the upper limit of algorithm iteration. Only by breaking through scarcity can continuous optimization be supported. Finally, the lack of a unified migration benchmark makes it impossible for different solutions to be truly compared, thereby hindering industry consensus and technological accumulation. In other words, generalization, data availability and evaluation criteria form an interdependent whole. Only when they are all stable can cross-domain prediction move from local trials to large-scale implementation.

6.2. Prospects for the future

Facing the next stage of technological transition, bearing RUL prediction urgently needs a research route from algorithm reliability to industrial-level iterative evolution. Firstly, it uses a rigorous quantitative framework to test the applicable boundary between attention and other time series mechanisms, and lays a credible cornerstone for the model. Then, multi-layer interpretation constraints are introduced to deepen the readability of the decision process. On this basis, efficient deployment in resource-constrained scenarios was realized through the collaboration of extremely lightweight and hybrid architecture. At the same time, relying on cloud edge joint and uninterrupted data flow, create a self-evolving real-time prediction system; Finally, with the help of model embedding, the problem of generalization across working conditions was overcome, and sustainable kinetic energy was provided for large-scale industrial landing.

(1) Evaluation of the applicability of attention mechanisms

Future research can systematically evaluate the applicability of attention mechanisms in RUL prediction from three perspectives. First, from the perspective of quantitative metrics, visualization techniques such as heatmaps and attention weight distribution can be used to verify whether the model focuses on key feature regions. Feature contribution analysis can help assess the relative importance of different inputs after attention weighting, while entropy or variance of attention distribution can indicate whether there is an issue such as mode collapse. Second, from the perspective of unsupervised or contrastive learning, methods like clustering or dimensionality reduction can be employed to check whether attention distributions vary between classes, and frameworks such as SimCLR (Chen et al., 2020) can be used to test the consistency and robustness of attention-based feature extraction under perturbations like time shifts or scaling. The principle is shown in Fig. 18.

Here,two separate data augmentation operators are sampled from the same family of augmentations $(t \sim \tau)$ and $(t' \sim \tau)$ and applied to each data example to obtain two correlated views. A base encoder network $f(\cdot)$ and a projection head $g(\cdot)$ are trained to maximize agreement using a contrastive loss.

Lastly, from the computational cost perspective, key metrics such as inference time and memory usage should be measured to determine whether the attention mechanism is suitable for deployment in resource-constrained environments, and whether its benefits in accuracy or interpretability justify the added cost in real-world RUL prediction tasks.

(2) Designing Methods to improve model interpretability

Physical prior knowledge can help explain the rationale behind model predictions, and future research may leverage this by incorporating physics-based constraints to guide attention mechanisms. One approach is to design physics-constrained networks or embed physical rules into the loss function, ensuring that the attention mechanism adheres to known physical laws during the learning process. Additionally, feature guidance strategies based on physical priors can be adopted to

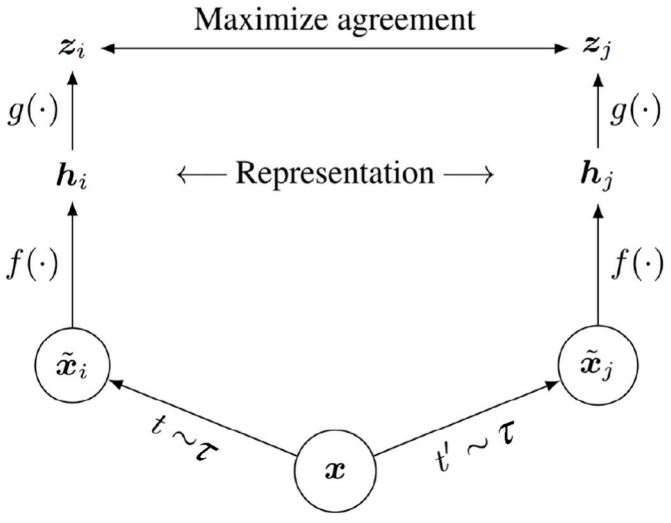


Fig. 18. A simple framework for contrastive learning of visual representations (Chen et al., 2020).

steer the model's focus toward features with clear physical meaning, thereby enhancing the reliability of predictions. Moreover, combining the attention mechanism with causal inference frameworks can help analyze the causal relationships between input features and model decisions. By modeling causal graphs, researchers can characterize the dependencies among different input features and quantify causal effects using causal discovery algorithms.

Finally, graph-based time series modeling methods, such as GCN, offer promising pathways to improve model interpretability. By calculating temporal similarity through methods like Dynamic Time Warping, a temporal graph can be constructed, enabling the GCN to learn and represent time dependencies more effectively and transparently. (Scarselli et al., 2009; Xiao et al., 2024).

(3) Optimization of deployment of lightweight time series and their hybrid models in industry

In lightweight deployment, model pruning and knowledge distillation are often used independently, but future research may explore their combined application to maximize compression efficiency(LeCun et al., n.d.). One promising direction is structural distillation, where pruning constraints guide the distillation process, allowing the student model to not only learn the knowledge from the teacher model but also automatically optimize its network architecture to eliminate redundant computation.

Notably, LNN offer a novel approach for lightweight and real-time prediction, Its specific framework is shown in Fig. 19. Based on continuous-time dynamical system models, LNN feature an internal "liquid layer" mechanism that can flexibly capture and exploit complex temporal dependencies within input sequences.

Compared to large-scale Transformer-based models, LNN can achieve comparable prediction accuracy with a dramatically smaller number of parameters, making them highly practical for industrial deployment, especially in resource-constrained environments (Chahine et al., 2023; Hasani et al., 2022).

(4) Continuous Iteration and Optimization of Online real-time prediction

Future research can further advance online real-time RUL prediction through continuous iteration and optimization in three key areas: data processing, model adaptability assessment, and computational efficiency enhancement. In terms of data processing, strategies such as dynamic sliding windows or variable-length windows can be explored to accommodate the temporal dynamics of bearing degradation processes. These can be combined with adaptive sampling rate mechanisms to adjust data acquisition frequency in real time, ensuring a balance between information richness and computational cost. For model adaptability assessment, uncertainty-based incremental learning methods can be used to dynamically determine whether the model needs to be updated, thereby reducing unnecessary computations.

When paired with knowledge distillation, such methods can help the model retain prior knowledge while learning from new data, mitigating the issue of catastrophic forgetting and improving the model's online adaptation capability. In the area of computational efficiency, techniques like pruning and quantization can reduce the inference burden. Additionally, combining edge computing for real-time deployment with cloud computing for heavy model optimization enables an efficient division of tasks-allowing terminal devices to execute lightweight, real-time predictions while leveraging cloud resources to optimize model complexity. This hybrid framework enhances both the stability and efficiency of real-time RUL prediction systems.

(5) Improve the generalization ability of the model by embedding other models

To enhance the robustness of bearing RUL prediction models under varying working conditions, future research can systematically explore data-centric approaches. By applying adversarial training or domain adaptation to vibration data from multiple operating conditions, models can become more resilient to distributional shifts. Semi-supervised or self-supervised learning methods can be used to leverage a small amount of labeled data alongside large volumes of unlabeled data, improving adaptability to new scenarios. Digital twin technology can generate synthetic data that closely resembles real-world conditions, enabling models to learn a wider range of potential failure patterns. Data augmentation and multimodal fusion techniques can be employed to extract consistent degradation features across sensors from different perspectives, addressing the challenge of limited sample sizes.

To fundamentally improve generalization and robustness, efforts should focus on developing standardized benchmarks and evaluation protocols, integrating data from multiple platforms, and building reproducible public datasets to support academic progress in RUL prediction.

Additionally, LLM can be embedded into neural architectures by transforming vibration, temperature, and rotational speed time series

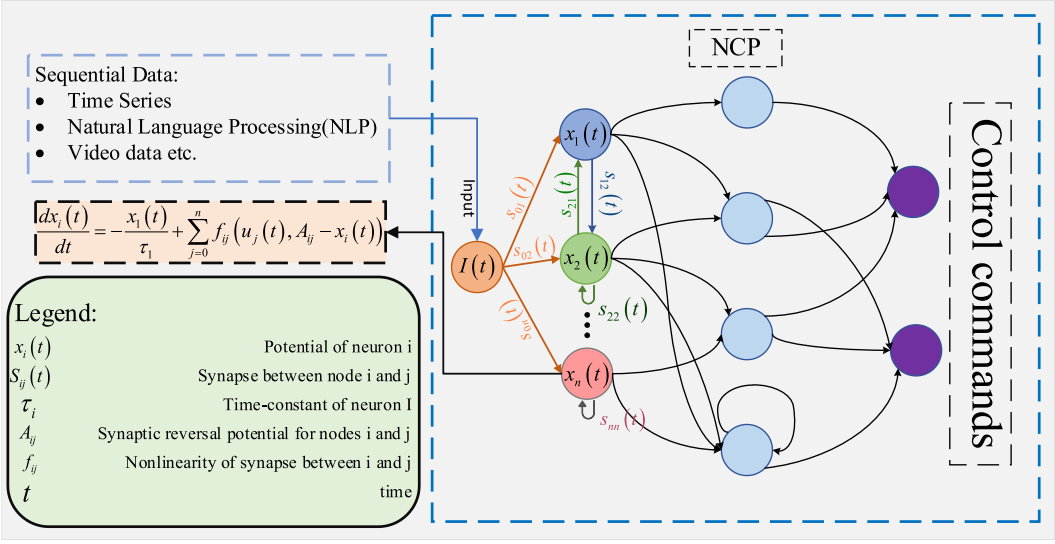


Fig. 19. A simple framework for LNN.

data into pseudo-language sequences. These can then be fed into pre-trained language models to uncover deep temporal relationships and hidden patterns, generating more expressive feature representations and significantly boosting the model's generalization capability. future research may also explore the integration of federated learning and reinforcement learning frameworks. Federated learning enables collaborative model training across distributed devices or institutions without sharing raw data, thereby preserving data privacy while leveraging diverse data sources to improve the model's cross-domain adaptability. On the other hand, reinforcement learning offers the potential to dynamically optimize maintenance decision-making by learning policies through interaction with the environment. When combined with RUL prediction, it can support adaptive maintenance strategies under uncertainty, further enhancing system-level intelligence.

7. Conclusion

This review systematically summarizes the research progress of DL-based time series forecasting models and hybrid models in bearing RUL prediction, analyzing their critical roles in improving prediction accuracy, enhancing model adaptability, and optimizing computational efficiency. It outlines the fundamental pipeline of RUL prediction-including data acquisition and preprocessing, feature extraction, model training, and performance evaluation-thus providing essential background and theoretical support for subsequent discussions.

(1) Through a comprehensive comparison of RNN, TCN, Transformer, and hybrid network strategies in bearing RUL prediction tasks, it is evident that model selection should be aligned with specific task objectives and balanced across multiple dimensions. While ensuring real-time feasibility, maximize the depth of abstraction and interpretation of the degradation mechanism. Looking to the future, the breakthrough point of research may not lie in simply stacking deeper networks, but in finding a new balance among cross-source data fusion, privacy-protected collaborative training, and decision-making closed-loop optimization.
(2) Future life prediction will no longer rely on a static "data-model-decision" pipeline, but evolve into a cross-layer adaptive intelligent ecosystem: On the data side, real-time perception and self-learning sampling are used to continuously correct the input. On the model side, physical-statistical fusion and federated continuous learning are used to self-iterate among multiple domains. The decision-making phase is shifting from static life prediction to comprehensive judgment oriented to operation and maintenance objectives, and strives to achieve a dynamic balance among maintenance cost, system risk and resource scheduling.
(3) The future prediction of bearing RUL should focus on three aspects: first, the credibility of attention mechanisms should be quantified by using physical consistency as a scale to lay a reliable base for all subsequent strategies; The second is to achieve millisecond efficient reasoning in resource-constrained scenarios through interleaved explanatory constraints, extreme lightweight and cloud-edge collaborative self-evolution. Third, we rely on self-supervised transfer, digital twin and federated learning to create a robust model ecosystem that can be replicated across working conditions. The above routes are coupled with each other and together point to a comprehensive transition from laboratory algorithms to industrial-grade health management.

This review provides a systematic analysis of the role of DL-based time series forecasting models in bearing RUL prediction and offers theoretical insight and practical guidance for continued innovation in this field.

This review is limited by potential selection bias in literature inclusion, as studies were chosen based on relevance and recent impact, possibly overlooking earlier significant contributions or emerging niche methodologies.

CRediT authorship contribution statement

Zhenyu Tang: Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Dalian Yang:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Linrong Tan:** Writing – review & editing, Investigation, Funding acquisition. **Liying Zeng:** Writing – review & editing, Investigation, Funding acquisition.

Data availability statement

No data was used for the research described in the article.

Ethical and informed consent for data used

This is a review article that does not involve ethical approval. No data sets were generated or analyzed in the article, and all the authors of the article were informed and agreed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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